The Effects of Monetary Policy in an Emerging Market Economy: An External Instrument Approach*

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Abstract

This paper analyzes the effects of monetary policy on the Mexican economy using a VAR model with external instruments. Monetary policy shocks are identified through high-frequency changes in interest rates around central bank announcements. The results show that a restrictive monetary policy shock leads to a decline in real output, which follows a U-shaped response and reaches its maximum effect after a lag. Importantly, using this identification strategy, both the price and exchange rate puzzles disappear, yielding results consistent with economic theory. Prices decline following a restrictive monetary policy shock, with the maximum effect occurring after a lag. The Mexican peso initially appreciates against the US dollar after the shock and then gradually depreciates back to its original level. The findings highlight the importance of the exchange rate channel in the transmission of monetary policy in Mexico, as it amplifies the response of prices.

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1 Introduction

Assessing the timing and effects of monetary policy on the economy is critical for central banks. Since the seminal work of Sims (1980), Vector Autoregression (VAR) models have been extensively employed to analyze this subject. However, the standard identification strategy in these models often involves imposing timing restrictions on the impact of the monetary policy rate. Common restrictions assume that within a period, the monetary policy rate responds to all other variables in the VAR model, but not vice versa. While these recursive identification strategies may be reasonable for interactions between the monetary policy rate and variables like output and prices, they imply challenges when financial variables such as the exchange rate are involved (Bjørnland, 2009; Capistrán et al., 2019). The issue lies in simultaneity: within a period, monetary policy shifts not only influence financial variables but may also be influenced by them.

Several empirical studies imposing timing restrictions in VAR models have reported a short-term increase in prices, commonly referred to as a "price puzzle", following an unexpected tightening of monetary policy.[1] This result, which is inconsistent with conventional macro models, raises questions about the accuracy of VAR models in identifying monetary policy shocks or the effectiveness of central banks in controlling inflation (Rusnak et al., 2013). Moreover, empirical evidence from open economies has also found puzzling results regarding the behavior of the exchange rate, observing a depreciation of the domestic currency after a contractionary monetary policy shock (Racette and Raynauld, 1992; Grilli and Roubini, 1995; Kim and Lim, 2022). In other instances, a persistent hump-shaped appreciation of the currency has been reported (Eichenbaum and Evans, 1995; Grilli and Roubini, 1995; Cushman and Zha, 1997; Lindé, 2003; Kim et al., 2017). These "exchange rate puzzles", more pronounced in emerging market economies (EMEs) compared to advanced

[1] According to the survey by Rusnak et al., (2013), approximately half of the studies employing VAR models to examine the effects of monetary policy shocks find that prices actually increase after an unexpected tightening, at least in the short run. Among the most common solutions to this puzzle that have been proposed in the literature are the inclusion of commodity prices or intermediate goods price measures as additional variables in the model and the adoption of alternative identification strategies, such as sign restrictions [see Canova and De Nicolo, 2002; Carrillo and Elizondo, 2015; Christiano et al., 2005; Uhlig, 2005; Sims, 1992; Sims and Zha, 2006].

economies (AEs) (Kim and Lim, 2022), contradict the overshooting hypothesis of Dornbusch (1976), which suggests that an unexpected increase in domestic short-term interest rates should trigger an instantaneous appreciation of the nominal exchange rate, followed by a gradual depreciation in line with the uncovered interest parity (UIP) condition (Bjørnland, 2009; Capistrán et al., 2019).[2]

This paper examines the effects of monetary policy in an EME, specifically Mexico, over the period 2004-2019.[3] For this purpose, we estimate a VAR model incorporating monetary policy surprises in a narrow window around central bank announcements in Mexico as an external instrument to identify monetary policy shocks. Through this model, we obtain the dynamic responses of key macroeconomic variables, including output, prices, and the exchange rate, to a contractionary monetary policy shock. This approach, which combines the strengths of traditional VAR analysis with high-frequency identification (HFI) methodologies, has proven useful for assessing the transmission of monetary policy in AEs.[4] We compare our baseline model with a recursive identification, an approach that has been widely used for Mexico (Sidaoui and Ramos-Francia, 2008; Capistrán et al., 2012; Ibarra, 2016; Chiguil-Rojas et al., 2024). Unlike the recursive identification scheme, the external instruments approach does not require imposing timing restrictions on the variables under analysis to identify the monetary policy shock. This is particularly crucial in the context of EMEs,

[2]The UIP condition stems from the principle of arbitrage, wherein investors seek to profit from differences in interest rates between countries. If the UIP condition holds, arbitrage activities should lead to adjustments in exchange rates that eliminate any potential profit from interest rate differentials.

[3]Mexico is a small open economy that has strong economic links with the US, often considered the "center country" in studies involving small economies. Approximately 80% of Mexican exports are directed to the US, while nearly half of the foreign direct investment received by Mexico originates from its northern neighbor (Carrillo et al., 2020). Among a group of large EMEs including Brazil, Chile, Malaysia, and South Africa, among others, Mexico has been characterized by the strongest trade relations with the US (Fink and Schüler, 2015). Furthermore, there exists significant financial integration between the Mexican and US economies. Variables such as the cost of external finance and the term premium show a correlation of around 75% and 67%, respectively. Additionally, short-term nominal interest rates in both countries exhibit a pronounced co-movement, displaying a correlation of 68% (Carrillo et al., 2020).

[4]The literature on HFI is large and expanding. Some of the early empirical studies assessing the impact of monetary policy surprises around central bank announcements on asset prices include Kuttner (2001), Gürkaynak et al. (2005) and Bernanke and Kuttner (2005), among others. Subsequent studies, including those by Gertler and Karadi (2015), Caldara and Herbst (2019), and Ha and So (2023), have integrated the HFI approach with VAR models to assess the effects of monetary policy on macroeconomic variables in AEs such as the US and Canada.

where imposing such restrictions has often led to puzzling outcomes involving both prices and the exchange rate that are difficult to reconcile with conventional economic theory.[5]

In line with Andrade and Ferroni (2021) and Solís (2023a, 2023b), we identify exogenous monetary policy surprises originating solely from central bank communication by examining intraday changes in the 3-month swap rate within a half-hour window around monetary policy announcements in Mexico.[6] Under the HFI approach, surprises in interest rates during monetary policy announcements are considered orthogonal to within-period fluctuations in both macroeconomic and financial variables. The fundamental assumption is that intraday changes in interest rates are solely influenced by monetary policy shocks. Thus, it is unlikely that shocks unrelated to these announcements systematically occur at the same time. This assumption is justified given that such changes are measured in a narrow time window around central bank announcements.[7]

As a by-product of our analysis, we can then explore the transmission of monetary policy to the Mexican economy through the exchange rate channel. We conduct this examination by comparing the responses of output and prices to a contractionary monetary shock in two identical VAR models, differing only in that one model blocks the exchange rate responses.[8] This

[5]Recent work argues that part of these puzzles may reflect central bank information shocks, arising when monetary policy announcements simultaneously convey interest rate decisions and the central bank's assessment of the economic outlook rather than a pure policy tightening (Jarociński and Karadi, 2020; Miranda-Agrippino and Ricco, 2021). In this view, surprise rate increases associated with positive news about the economy can generate co-movements in prices, asset prices and output that resemble the so-called puzzles. However, the precise mechanism behind these co-movements remains subject to debate. Bauer and Swanson (2023a, 2023b), for instance, argue that a "Fed response to news" can induce similar biases when the central bank reacts more aggressively to public information than markets anticipate. In addition, the presence and magnitude of such information effects appears to vary across countries, samples and identification schemes. For example, Cesa-Bianchi et al. (2020) for the United Kingdom and Soosalu (2024) for Canada find no evidence that information effects contaminate their identification of monetary policy surprises. For small open economies such as Mexico, evidence is still scarce, which motivates our discussion of their potential relevance for our identification strategy in Section 3.3.3, where we assess whether our results are preserved when we account for these central bank information shocks.

[6]These changes capture surprises not only about the current policy rate level but also about the expectations of its future path in the short run, as 3-month swap contracts may cover more than one policy meeting ahead (Solís, 2023a; 2023b).

[7]This is a standard assumption in the literature on HFI (Gürkaynak et al., 2005).

[8]Morsink and Bayoumi (2001), Disyatat and Vongsinsirikul (2003), Aleem (2010), and Ibarra (2016) have employed a similar methodology to examine different monetary policy transmission channels in Japan, Thailand, India, and Mexico, respectively. In particular, Ibarra (2016) examines how important is the credit channel in the transmission of monetary policy in Mexico.

channel holds particular significance for open economies like Mexico (Solís, 2023a; 2023b), which heavily rely on international trade. Adjustments in the exchange rate can significantly affect export competitiveness, import prices, and consequently, overall economic conditions such as economic activity and inflation. A recent empirical study by Capistrán et al. (2019) suggests that one of the primary mechanisms through which monetary policy influences inflation in Mexico is closely linked to its short-term impact on the exchange rate. This may be attributed to the increased sensitivity of the exchange rate to monetary policy news, particularly evident after the global financial crisis (Ferrari et al., 2021). Following this period, central banks worldwide have sought to enhance transparency and public communication in monetary policy (Trung, 2022). Notably, unconventional monetary policies (UMPs), particularly those related to communication such as forward guidance, have played a significant role in driving exchange rate movements (Ferrari et al., 2021), potentially amplifying the exchange rate channel as a transmission mechanism.[9,10]

The main results indicate that monetary policy shocks, identified using high-frequency surprises around central bank announcements in Mexico, lead to responses in output, prices, and the exchange rate that align closely with conventional macroeconomic models. Importantly, we show that by employing the external instruments identification strategy, both the so-called price and exchange rate puzzles disappear. Following a restrictive monetary policy shock, standardized as a one percentage point increase in the one-year interest rate, Mexican output displays a U-shaped response, reaching its lowest point in the tenth month with a

[9]For a more complete description of these policies, see, for example, Moessner et al. (2017) and Lombardi et al. (2018). In general terms, UMPs can be defined as any policy, other than the setting of short-term interest rates, that aims at achieving a stated monetary policy objective either by influencing economic outcomes or by moderating shocks to the financial system (Lombardi et al., 2018). These policies, in turn, can be introduced in the form of quantitative easing (QE) or forward guidance. While the former policies may involve direct intervention in the monetary system through credit policies or asset purchase programs, for instance, the latter ones aim to change expectations by sending signals about the future policy path or the macroeconomic outlook in policy statements; that is, forward guidance policies use communication to affect policy outcomes (Campbell et al., 2012; Moessner et al., 2017; Lombardi et al., 2018).

[10]Central banks' transparent communication and implementation of forward guidance can enhance the efficacy of monetary policy, fostering greater confidence and reducing uncertainty about future monetary policy decisions (Blinder et al., 2008). For the particular case of Mexico, Solis (2023b) shows that the central bank conducts monetary policy by adjusting its current policy rate and by managing expectations about its future path via statements, both of which strongly influence the exchange rate.

decline ranging between 0.55% and 0.83% below the baseline. In turn, prices also decrease in response to the aforementioned shock. The maximum decline in this variable is observed 18 months after the shock, ranging between 0.16% and 0.45% below the baseline. These results are consistent with the average findings of cross-country survey studies by Havranek and Rusnak (2018) and Nguyen (2020).[11] Regarding the response of the exchange rate to a contractionary monetary policy shock, the Mexican peso immediately appreciates against the US dollar by approximately 2.53% at the median, followed by a gradual depreciation until it returns to the original level. This response, in line with Bjørnland (2009), Capistrán et al., (2019), and Ha and So (2020), aligns with the predictions of the overshooting theory proposed by Dornbusch (1976).

Regarding the relevance of the exchange rate channel as a transmission mechanism of monetary policy in Mexico, our findings suggest that it seems to explain an important part of the responses of both output and prices to a contractionary monetary policy shock. In particular, allowing this channel to operate leads to a notably faster and stronger price response. Additionally, real output reacts more rapidly to the aforementioned shock, although the maximum decrease in this variable is very similar to the maximum decrease observed when the exchange rate channel is not allowed to operate. These results are consistent with Capistrán et al. (2019) and may suggest that an important mechanism through which the exchange rate influences inflation in Mexico could be fundamentally linked to its direct effects on the price level or on inflation expectations. Specifically, the strong appreciation of the Mexican peso in response to a contractionary monetary policy shock seems to lead to a decrease in the prices of imported goods, contributing to a reduction in inflation. Furthermore, the appreciation of

[11] The results are also consistent with those of Deb et al. (2023) for emerging economies in terms of the timing of monetary policy effects, which also peak within two years. While the magnitude of the inflation response is comparable to their estimates, the output effects in our case appear to be larger. This difference may reflect the way the monetary policy shock is standardized. In our case, it corresponds to a one percentage point increase in the one-year interest rate, whereas Deb et al. (2023) define it as a one percentage point increase in short-term interest rates. The shock to the one-year interest rate is expected to generate larger effects, as it also incorporates changes in expectations regarding future monetary policy. In addition, since the pass-through from the policy rate to the one-year rate is incomplete, our shock represents a stronger monetary tightening. More broadly, studies using high-frequency identification strategies that standardize the shock as a one percentage point increase in the one-year rate tend to find stronger effects on both output and inflation than traditional approaches. For example, Checo et al. (2024) report output responses of similar magnitude to ours and even larger inflation responses.

the domestic currency could prompt a downward adjustment in inflation expectations, thereby exerting further downward pressure on inflationary pressures. In terms of output, the appreciation of the domestic currency reduces export competitiveness and makes imports more attractive, leading to a decline in net exports. This contraction in external demand contributes to the observed decrease in real output.

This paper contributes to the literature in two main dimensions. First, it represents one of the initial attempts to analyze the effects of monetary policy in an EME using an external instruments approach, specifically employing high-frequency surprises extracted from financial market data around monetary policy announcements. We focus on Mexico as a fine example of an EME for this study, adopting a country-specific design in contrast to existing cross-country panel evidence [see Deb et al., 2023, and Checo et al., 2024]. This economy has relatively liquid financial markets, a floating exchange rate, and a credible inflation targeting regime since 2001, making it a suitable candidate for this type of analysis.[12] The increasing use of the external instruments approach in the field of monetary economics, particularly within the framework of AEs, serves as motivation for its application in EMEs, where the transmission of monetary policy is less conclusive (Nguyen, 2020).[13] As mentioned above, a major challenge in analyzing this subject using VAR models is how to address simultaneity issues among the variables involved in the analysis. As in Gertler and Karadi (2015), the empirical strategy employed in this study addresses simultaneity by using an external instrument instead of relying on assumptions about causal relationships among endogenous variables. By employing this identification strategy, we contribute to the VAR literature for EMEs by showing that it constitutes one way to resolve both the price and exchange rate puzzles. Second, this study examines the transmission of monetary policy in the Mexican

[12] One critical reason for the lack of studies for EMEs may be the absence of futures markets with active trading for the operating targets of monetary policy in those economies, making high-frequency identification of monetary surprises not easily applicable.

[13] Havranek and Rusnak (2018) and Nguyen (2020) provide insights into the literature, offering estimates on the average lag length and the sources of variability in the transmission of monetary policy in these economies. Havranek and Rusnak (2018) collect sixty-seven published studies, encompassing 12 advanced and emerging market economies, to examine when prices bottom out after a monetary policy contraction. Nguyen (2020), on the other hand, synthesizes findings on the output effect of monetary policy shocks in 32 developing and emerging economies.

economy through the exchange rate channel. Understanding how the central bank influences the economy through this channel is crucial for the design of monetary policy in EMEs. Moreover, analyzing the exchange rate channel contributes to the ongoing discussion regarding its importance for the Mexican economy.

Related literature. This paper is related to a broad body of literature measuring the effectiveness of monetary policy in EMEs. The evidence generally points to contractionary effects on activity and prices, with timing and magnitudes varying across countries and regimes. For Latin America, several studies document that output typically reaches its peak within about a year, while inflation declines over horizons of roughly two years (Minella, 2003; Céspedes et al., 2008; Ormaechea and Fernandez, 2011). In an analysis for Chile, Brazil, Colombia, Peru, and Mexico, Otero (2015) finds larger real and nominal responses in Mexico and Peru relative to peers. Evidence for other EMEs, such as Brazil, Russia, India, China, and South Africa (BRICS) suggests that a tightening reduces GDP by about 0.2% within four quarters, with inflation falling quickly but only temporarily (Mallick and Sousa, 2012). We contribute to this literature by providing evidence for Mexico, a fine example of an EME, using an external instruments approach to identify monetary policy shocks.[14]

Our work is also related to the recent literature that relies on high-frequency surprises as external instruments to identify monetary policy shocks. For the US, Gertler and Karadi (2015) employ intraday interest rate futures surprises as instruments and show that a contractionary policy shock generates responses in real activity and the price level that are consistent with standard evidence. Cesa-Bianchi et al. (2020) build a series of high-frequency interest rate surprises for the UK and, using them as an external instrument, document contractionary effects on activity and prices, as well as an appreciation of the exchange rate. They also find that their monetary policy surprises are not contaminated by information effects. For Canada, Soosalu (2024) constructs surprises from Bankers' Acceptance futures and shows

[14] Recent evidence suggests that the Mexican economy has relatively liquid financial markets. For instance, Ibarra (2023) indicates that, over the last two decades, Mexico has had the highest investment in government securities by non-residents among emerging economies, while a survey by the Emerging Markets Traders Association reports that in 2025 Mexican debt instruments were the most frequently traded among emerging markets (Murno, 2025).

that a 25 basis point contractionary shock reduces output by about 0.5 percentage points after 18 months and lowers the price level by around 0.3 percentage points after two years, and also points out that their surprise measure is robust to information effects. We build on this literature by providing further evidence for a small open economy, such as Mexico.

Recent contributions use external instruments in panel settings, but with shocks that differ from high-frequency surprises. For instance, Checo et al. (2024) analyze the transmission of monetary policy by constructing a novel data set for 18 EMEs from Bloomberg analysts' forecast errors of policy rate decisions. They find that a 100 basis point surprise tightening raises bond yields, appreciates the exchange rate, and reduces industrial production for several quarters, while inflation declines more slowly. Deb et al. (2023) also study the monetary policy transmission for 33 advanced and emerging economies using a novel dataset of monetary policy shocks constructed from Consensus Forecasts errors in short-term interest rates. They show that a 100 basis point rate hike lowers real GDP by about 0.3 percent within a year, with relatively muted and delayed inflation responses in EMEs. While these studies provide valuable cross-country evidence on monetary policy transmission in EMEs, our paper differs by focusing on a single emerging economy, Mexico, and by using an external instrument based on high-frequency financial data instead of survey-based data, which allows us to obtain country-specific impulse responses and to quantify transmission through the exchange rate channel.

Moreover, our work also relates to the body of literature on the monetary policy transmission in Mexico, particularly analyzing the exchange rate channel. Early contributions document an operative interest rate, credit and exchange rate channels (Copelman and Werner, 1997; Díaz de León and Greenham, 2001; Martínez et al., 2001). Subsequent studies around the adoption of inflation targeting in Mexico emphasize structural changes, such as a lower exchange rate pass-through, stronger interest rate responses, and a greater role for forward-looking inflation dynamics (Gaytán and González-García, 2007; Sidaoui and Ramos-Francia, 2008; Ramos-Francia and Torres, 2008; Capistrán et al., 2012; Cortés, 2013). More recent studies highlight a rapid exchange rate channel consistent with the overshooting hypothesis (Capistrán et al., 2019). Our paper differs from these studies by employing an external instrument approach based on high-frequency financial data, and by examining how the transmission would change if the exchange rate channel were shut down.

The remainder of this paper is organized as follows. Section 2 describes the VAR model and the data used in the estimation. The estimation results are reported and discussed in Section 3. The last section concludes and discusses areas for future research.

2 Methodology

2.1 A VAR Model for a Small Open Economy

The reduced form representation of the VAR model is as follows:

$$\boldsymbol{Y_t} = \boldsymbol{c} + \sum_{j=1}^{p} \boldsymbol{B_j} \boldsymbol{Y}_{t-j} + \boldsymbol{\delta} \boldsymbol{X}_t + \boldsymbol{u_t}, \tag{1}$$

where $\boldsymbol{Y}_t$ represents a vector of domestic macroeconomic and financial variables, including a monetary policy indicator, $\boldsymbol{c}$ is a vector of constants, $\boldsymbol{B_j}$ denotes the coefficient matrix at lag j, $\boldsymbol{X}_t$ is a vector of exogenous foreign variables, mainly from the US, and $\boldsymbol{u}_t \sim N(0, \boldsymbol{\Sigma})$ is a vector of white noise residuals that are normally distributed. As in Gertler and Karadi (2015), we differentiate between the monetary policy indicator and the monetary policy rate, which are typically considered the same in the standard VAR model. In particular, we use a government bond rate with a maturity longer than the current period monetary policy rate as the policy indicator. The advantage of using the government bond rate is that its innovations encompass not only the effects of surprises in the current monetary policy rate but also changes in expectations about the future trajectory of this rate.[15] Following the approach commonly used in the literature for small open economies, we include exogenous foreign variables primarily from the US to capture international economic events. These exogenous foreign variables are assumed to have contemporaneous effects on the endogenous domestic variables, but there is no feedback effect from the domestic variables to the foreign vari-

[15] Further details regarding our monetary policy indicator are provided in Section 2.3.

ables. This assumption is common in the structural VAR literature examining the impacts of monetary policy in EMEs (Bjørnland, 2009; Ibarra, 2016; Capistrán et al., 2019; Kim and Lim, 2022). The reduced-form residuals are given by the following function of the structural shocks:

$$\boldsymbol{u_t} = \boldsymbol{S\epsilon_t} \tag{2}$$

Here, $\boldsymbol{S}$ is a nonsingular square matrix, which is equivalent to assuming that the VAR is invertible [see Stock and Watson (2018)]. Let $\boldsymbol{s}$ denote the $n \times 1$ column in matrix $\boldsymbol{S}$ corresponding to the impact effects of the structural monetary policy shock ε_t^p on $\boldsymbol{u}_t$ and, consequently, on $\boldsymbol{Y_t}$. Therefore, to compute the impulse response functions to a monetary policy shock, we need to estimate:

$$\boldsymbol{Y_t} = \boldsymbol{c} + \sum_{j=1}^{p} \boldsymbol{B_j} \boldsymbol{Y}_{t-j} + \boldsymbol{\delta X}_t + \boldsymbol{s}\varepsilon_t^p \tag{3}$$

Given that recursive identification has been a widely used approach in the Mexican monetary policy literature, we also adopt this scheme as a comparison identification strategy [see Sidaoui and Ramos-Francia, 2008; Capistrán et al., 2012; Ibarra, 2016; Chiguil-Rojas et al., 2024]. In the standard recursive identification approach, it is typically assumed that all elements of the vector $\boldsymbol{s}$ are zero except for the one corresponding to the monetary policy indicator. While this assumption helps in identifying $\boldsymbol{s}$, it becomes problematic when financial variables such as the exchange rate are included in the VAR model. Imposing a restriction that an innovation in the monetary policy indicator has no contemporaneous impact on other financial variables is often implausible.[16] The other option is to impose the restriction that financial variables have no contemporaneous impact on the monetary policy indicator. However, it is also challenging to argue that current monetary policy does not respond to information contained in financial variables (Gertler and Karadi, 2015). Given our interest in examining the joint response of economic and financial variables, an alternative approach to

[16] There is considerable evidence from the HFI literature indicating that financial variables respond immediately and substantively to an unexpected monetary policy tightening [see, for instance, Andersen et al. (2003, 2007); Ehrmann and Fratzscher (2003, 2005); Faust et al. (2007); Bredin et al. (2010); Hausman and Wongswan (2011); Wright (2012); Rogers et al. (2014); Solís (2023a, 2023b)].

identifying monetary policy shocks is necessary. Thus, we opt to use external instruments as our identification strategy.[17]

2.2 Identification: The External Instruments Approach

Following Stock and Watson (2012), Mertens and Ravn (2013), and Gertler and Karadi (2015), let $\widetilde{\boldsymbol{Z}}_t$ denote the series of high-frequency changes in our monetary policy external instrument around all central bank announcements, and let $\boldsymbol{Z}_t$denote the corresponding monthly version. Because announcements occur on different days within the month and the VAR is estimated at a monthly frequency, we map $\widetilde{\boldsymbol{Z}}_t$ into $\boldsymbol{Z}_t$ by averaging within-month cumulative surprises and then taking first differences to obtain monthly surprises (details are provided in Section 2.3). In addition, let ε_t^q represent a vector including structural shocks different from the monetary policy shock ε_t^p. For $\boldsymbol{Z}_t$ to be a valid instrument for ε_t^p, it must satisfy an instrument relevance condition and an instrument exogeneity condition, as follows:

$$E[\boldsymbol{Z}_t \boldsymbol{\epsilon}_t^p] = \boldsymbol{\Phi} \tag{4}$$

$$E[\boldsymbol{Z}_t \boldsymbol{\epsilon}_t^q] = \mathbf{0} \tag{5}$$

The advantage of using high-frequency interest rate changes around monetary policy announcements lies in their potential to satisfy conditions (4) and (5). First, monetary policy announcements represent a crucial element of the news on monetary policy, suggesting a strong positive correlation between $\boldsymbol{Z}_t$ and ε_t^p as outlined in (4). Second, since $\boldsymbol{Z}_t$ encompasses high-frequency interest rate changes within narrow time windows surrounding cen-

[17] An alternative methodology for shock identification in structural VAR models is the use of sign restrictions (Uhlig, 2005). Rather than imposing a specific contemporaneous ordering of variables, this approach identifies structural shocks by restricting the sign (positive, negative, or zero) of the impulse responses of selected variables over a time horizon (typically on impact or during the initial periods). Only the impulse response functions that comply with the imposed restrictions are retained. This method is especially useful when economic theory or empirical evidence provides guidance on the qualitative effects of shocks, but not on their exact timing or magnitude. An important advantage of HFI over sign restrictions is its ability to isolate exogenous monetary policy shocks using actual market surprises observed in high-frequency financial data around policy announcements. By focusing on narrow time windows, HFI captures the unexpected component of policy actions in a more data-driven manner, without the need for theoretical restrictions on responses.

tral bank announcements, the likelihood of other structural shocks systematically influencing monetary policy decisions precisely at those moments is minimal (Kuttner, 2001; Gürkaynak et al., 2005). This implies an orthogonal relationship between $\boldsymbol{Z}_t$ and ε_t^q, as delineated in (5).

Let $\boldsymbol{u}_t^p$ represent the reduced-form residual from the VAR equation for the monetary policy indicator, $\boldsymbol{u}_t^q$ denote the reduced-form residual from the equation for variables $q \neq p$, and $\boldsymbol{s}^q \in \boldsymbol{s}$ represent the impact effect of ε_t^p on $\boldsymbol{u}_t^q$. To estimate the elements in the vector $\boldsymbol{s}$ in Eq. (3), we proceed as follows. We first obtain estimates of $\boldsymbol{u}_t$ via ordinary least squares. Then, we isolate the variation in $\boldsymbol{u}_t^p$ that is due to ε_t^p. This is achieved by regressing $\boldsymbol{u}_t^p$ on our external instrument $\boldsymbol{Z}_t$ and a constant to form the fitted value $\hat{\boldsymbol{u}}_t^p$. An important statistic for assessing the quality of $\boldsymbol{Z}_t$ in this step is its first-stage F-statistic. Specifically, it is desirable that the instrument has an associated F-statistic above the weak instruments threshold value of 10 suggested by Stock et al. (2002). Finally, we estimate the ratio $\boldsymbol{s}^q/s^p$ through a least squares regression of $\boldsymbol{u}_t^q$ on $\hat{\boldsymbol{u}}_t^p$ as follows:

$$\boldsymbol{u}_t^q = \frac{\boldsymbol{s}^q}{s^p}\hat{\boldsymbol{u}}_t^p + \boldsymbol{\xi}_t, \tag{6}$$

where $\hat{\boldsymbol{u}}_t^p$ is orthogonal to the error term $\boldsymbol{\xi}_t$, given the assumption of (5) that $\boldsymbol{Z_t}$ is orthogonal to all the structural shocks other than ε_t^p. Given that the variation in $\hat{\boldsymbol{u}}_t^p$ is solely attributable to ε_t^p, the second stage regression of $\boldsymbol{u}_t^q$ on $\hat{\boldsymbol{u}}_t^p$ then provides a consistent estimate of $\boldsymbol{s}^q/s^p$. An estimate for s^p is then derived using the estimated reduced form variance-covariance matrix $\boldsymbol{\Sigma}$ and Eq. (6). The details are provided in Appendix A. We are then able to identify $\boldsymbol{s^q}$. Given estimates of s^p, $\boldsymbol{s^q}$, $\boldsymbol{c}$, $\boldsymbol{B_j}$, and $\boldsymbol{\delta}$ we can use Eq. (3) to compute impulse response functions to monetary policy shocks.

Finally, we estimate confidence intervals using a residual-based moving block bootstrap (MBB). Unlike the wild bootstrap, MBB resamples blocks of residuals and the instrument in order to preserve their serial dependence and covariance, which are crucial for valid inference in proxy VARs, as emphasized by Jentsch and Lunsford (2019). We implement MBB with a block length of 72 months, which lies within the 18–96 month range used to define business cycle frequencies, as in Cuadra (2008). Consistent with standard practice in MBB inference for proxy VARs, we report 68% confidence intervals (Jentsh and Lunsford, 2019; Mertens and Ravn, 2019; Cesa-Bianchi et al., 2020).

2.3 Data Description

The high-frequency changes in our monetary policy instrument ($\widetilde{Z}_t$) around central bank announcements in Mexico are obtained from the dataset used in Solís (2023a), covering 72 scheduled announcements from January 2011 to December 2019. Following the usual approach in the related literature, these changes are computed as the variation in the 3-month swap rate within a 30-minute window, spanning from 10 minutes before to 20 minutes after the monetary policy announcements. Given that 3-month swap contracts may cover more than one policy meeting ahead, variations in the 3-month swap rate enable us to capture surprises not only regarding the current level of the monetary policy rate but also changes in expectations about its future path (Solís, 2023a; 2023b). Consequently, these surprises represent a comprehensive measure of the overall monetary policy stance adopted by Banco de México. The swap market in Mexico references an interbank interest rate closely tied to the monetary policy rate, known as the 28-day interbank interest rate. This rate, in turn, serves as the benchmark rate for banking loans in Mexico. The most liquid swap referencing the 28-day interbank interest rate, and indeed the primary derivative in the local market, is the 3-month swap (Solís, 2023a; 2023b).[18] Importantly, the 3-month swap is actively traded throughout the day, enabling the computation of high-frequency changes in the swap rate within 30-minute intervals.

Since the timing of monetary policy announcements varies throughout the month, and our VAR model employs monthly averages for financial variables, the impact of a surprise occurring at the end of a month may differ from one occurring at the beginning. To address this issue, we compute monthly average surprises, denoted as Z_t, following Romer and

[18] It would be informative to incorporate interest rate swap contracts with a longer maturity into our analysis. However, due to data unavailability, we confine the analysis to the 3-month swap contract. As a robustness check, however, we employ an alternative measure computed as the variation in the 3-month swap rate within a 50-minute window, spanning from 20 minutes before to 30 minutes after the announcement. The results are consistent with our baseline specification and are reported in Figures B.1 and B.2 of Appendix B.

Romer (2004), Barakchian and Crowe (2013), and Gertler and Karadi (2015). In particular, we proceed as follows. First, we generate a cumulative daily surprise series by aggregating all surprises observed on monetary policy announcement days. Next, we compute monthly averages of these cumulative surprise series. Finally, we derive monthly average surprises as the first difference of these series.

The variables included in the VAR model, along with their respective sources and transformations, are presented in Table 1. Given that output and price measurements reflect developments over the entire month rather than purely at a point in time, all low-frequency financial variables described in this table are included as monthly average figures.[19] The variables used in our analysis cover various categories, including foreign economic activity and prices, the money market, the debt market, commodities, country risk, foreign exchange, and domestic economic activity and prices. These variables were selected based on previous studies estimating VAR models for small open economies and assessing the impacts of monetary policy, such as Mackowiak (2007), Vicondoa (2019), Iacoviello and Navarro (2019), Capistrán et al. (2019), Carrillo et al. (2020), and Ha and So (2020), among others.

We estimate the VAR model in log levels, following standard practice in the monetary policy transmission literature (Bernanke and Blinder, 1992; Bernanke and Gertler, 1995; Christiano et al., 2005). This applies to real output, prices, and the exchange rate. Estimating in levels preserves long-run relationships and retains valuable information that would be lost through differencing (Sims et al., 1990). Although this may reduce estimation efficiency, our goal is to analyze dynamic relationships rather than optimizing forecasting performance. We also avoid using the output gap due to the substantial uncertainty surrounding its real-time estimation and frequent revisions (Orphanides, 2001).

In our group of exogenous variables, we include the shadow interest rate by Wu and Xia (2016), denoted as i_t^*, which allows us to account for the UMP implemented by the Federal Reserve in the wake of the global financial crisis. The shadow interest rate is essentially the federal funds rate during conventional monetary policy periods, while it reflects the effect of

[19]The exception is the shadow interest rate considering that data are just available as end-of-the-month figures.

Category	Series	Definition	Transformation	Source
		External instrument		
Money market	$\widetilde{Z}_t$	3-month swap rate surprise	No transformation	Solís (2023a)
		Exogenous variables contained in vector $\boldsymbol{X}_t$		
Money market	i_t^*	Shadow interest rate	No transformation	Wu and Xia (2016)
Economic activity	y_t^*	Interpolated measure of US GDP	$\log(y_t^*)$	Elizondo (2012)
Prices	p_t^*	Consumer Price Index	$\log(p_t^*)$	FRED
Commodities	wti_t	WTI crude oil price	$\log(wti_t)$	FRED
Debt market	$i_t^{*10y} - i_t^{*2y}$	US 10-2 year Treasury yield spread	No transformation	FRED
		Mexican variables contained in vector $\boldsymbol{Y}_t$		
Money market	i_t^{1y}	1-year government bond yield	No transformation	Banco de México
Economic activity	y_t	Interpolated measure of GDP	$\log(y_t)$	Elizondo (2012)
Prices	p_t	Consumer Price Index	$\log(p_t)$	INEGI
FX market	e_t	Exchange rate	$\log(e_t)$	Banco de México
Country risk	$embi_t$	Mexico's EMBI plus spread	No transformation	Bloomberg

Table 1: Data description

Note: The VAR variables are sampled from 2004M1–2019M12. The external instrument is available for 2011M1–2019M12.

expansionary monetary policies on the money market as it is not bounded by 0 percent.[20] Additionally, we incorporate an interpolated measure of GDP (y_t^*) obtained by Elizondo (2012) and the Consumer Price Index (p_t^*) as indicators of real output and the price level in the US, respectively.[21] The West Texas Intermediate (WTI) crude oil price is included due to its importance in determining Mexico's terms of trade (Capistrán et al., 2019).[22] Furthermore,

[20] In particular, the Federal Open Market Committee targeted the federal funds rate between 0 to 0.25 percent from December 16, 2008, to December 15, 2015. In this "zero lower bound" environment, some studies have used shadow rate models to characterize the term structure of interest rates or quantify the stance of monetary policy (Kim and Singleton, 2012; Bauer and Rudebusch, 2016; Krippner, 2013 and Wu and Xia, 2016). For robustness, we also estimate the model using the US two-year government bond rate instead of the shadow interest rate. As indicated by Gertler and Karadi (2015), Swanson and Williams (2014) and Hanson and Stein (2015), among others, such a rate allows us to account for the Federal Reserve's forward guidance strategy implemented after the global financial crisis, which seems to operate with a roughly two-year horizon. The results are consistent with our baseline specification and are reported in Figures B.3 and B.4 of Appendix B.

[21] Elizondo (2012) employs the Denton method to interpolate the GDP series utilizing the US Industrial Production Index. The estimated series closely aligns with the observed quarterly GDP. As a robustness exercise, we use the Industrial Production index as an alternative measure of economic activity in the US. The results reported in Figures B.5 and B.6 of Appendix B are consistent with our baseline specification.

[22] The inclusion of a commodity price variable is considered a "good practice" in the literature as it helps to mitigate the price puzzle often encountered in VAR models when using recursive identification approaches (Sims

we incorporate the yield spread between US 10-year and 2-year Treasury bond yields. This spread may aggregate high-quality forward-looking information that is relevant for the Mexican economy in the VAR model (Carrillo et al., 2020).[23] Finally, following Ha and So (2020), we include a dummy variable to account for the potential impacts of the global financial crisis on our estimates. This variable takes a value of 1 for the period spanning from July 2008 to June 2009 and 0 otherwise.[24]

Regarding Mexico's macroeconomic and financial variables, we employ the 1-year government bond yield i_t^{1y} as the monetary policy indicator. This rate not only reflects current monetary policy rate levels but also incorporates expectations regarding future policy adjustments (Gertler and Karadi, 2015).[25] Therefore, by employing i_t^{1y} alongside our monetary policy instrument, we can examine broader effects of monetary policy announcements on the economy. Additionally, we include an estimated monthly series for the real GDP (y_t) using the approach proposed by Elizondo (2012).[26] To measure domestic price levels, we use the Consumer Price Index p_t. This variable is seasonally adjusted with the X13-ARIMA method. Furthermore, we include the nominal peso-dollar exchange rate and Mexico's EMBI plus spread. The EMBI plus spread serves as an indicator of sovereign risk, representing the

and Zha, 2006; Christiano et al., 2005), which we also employ for comparison with the external instruments approach.

[23] Changes in the US yield spread may signal shifts in expectations regarding future monetary policy actions by the Federal Reserve and future economic conditions in the US. Since Mexico's economy is closely linked to the US economy through trade, investment, and financial channels, changes in US monetary and economic conditions may have significant spillover effects on Mexico.

[24] During this period the most severe impacts effects of the global financial crisis were observed on both AEs and EMEs such as Mexico (Banco de México, 2008). An alternative approach commonly used in the literature to construct a dummy variable of this nature involves identifying observations that exceed n times the interquartile range from the median for the entire sample (Stock and Watson, 2002). By setting $n = 2$ and examining outliers in economic activity variables, we find that the dummy variable essentially remains unchanged.

[25] Another commonly used monetary policy indicator in the literature is the two-year government bond rate. However, in line with Gertler and Karadi (2015), we find that surprises in the 3-month swap rate are not strong instruments for such a rate, but they are for the one-year interest rate. As a robustness check, we also show that our main results hold when using the two-year rate, albeit with the caveat that its use may imply a weak instruments problem. The results are reported in Figures B.7 and B.8 of Appendix B.

[26] Elizondo (2012) employs the Kalman filter to interpolate the monthly GDP series, utilizing the Overall Indicator of Economic Activity (IGAE by its Spanish acronym) available at a monthly frequency. IGAE follows the methodology and conceptual framework of the national accounts. Notably, the correlation between IGAE and GDP, both variables measured in quarterly percentage changes, is 0.99. As a robustness exercise, we use the IGAE index as an alternative measure of economic activity in Mexico. The results, reported in Figures B.5 and B.6 of Appendix B, are consistent with our baseline specification.

premium associated with long-term bonds issued by the Mexican government in US dollars (Carrillo et al., 2020). By including this variable, we account for changes in market perceptions of Mexico's sovereign creditworthiness, which may influence macroeconomic and financial conditions within the country.[27]

One advantage of the external instruments approach is its flexibility regarding the sample used for estimating the VAR model compared to the two-stage least squares regression used to estimate the vector s in Eq. (3). This flexibility is crucial because high-frequency interest rate data, commonly used in this approach, is typically available for a shorter period than the VAR data. In our case, the series of intraday changes in the 3-month swap rate are available from January 2011. Therefore, in line with Bauer and Swanson (2023b) and Gertler and Karadi (2015), we employ a longer sample period to estimate the VAR model, spanning from January 2004 to December 2019. Given the substantial number of parameters to be estimated, this allows us to avoid the common problems associated with short sample periods such as data overfitting, increased uncertainty, and unreliable parameter estimates, among others. However, to conduct the two-stage least squares regression and identify the vector $\boldsymbol{s}$, we use a shorter sample corresponding to the period from January 2011 to December 2019. This step involves only the estimated reduced-form residuals obtained from the VAR model and our external instrument.[28] We begin our sample in 2004 because it aligns with the start of the transition period for the adoption of the current monetary policy tool in Mexico, namely the overnight interbank interest rate, which our external instrument closely tracks (Solís, 2023a; 2023b).[29] In turn, the sample ends in 2019 to avoid the potential influence on our

[27] In particular, fluctuations in the EMBI plus spread can signal changes in global investor sentiment, liquidity conditions, and risk appetite, all of which may impact Mexico's economy and financial system. In the literature, the EMBI plus spread is also known as the country-interest-rate spread [see Uribe and Yue (2006)].

[28] One alternative to address the discrepancy between the samples used for estimating the VAR model and the two-stage least squares regression involves utilizing daily changes in the 3-month swap rate around monetary policy announcements as our external instrument, which can be computed from 2004. However, when considering the same sample period available for intraday changes in the 3-month swap rate, daily changes in this rate do not seem to serve as a suitable instrument for our monetary policy indicator, as indicated by the robust F-statistic falling far below the threshold value of 10 suggested by Stock et al. (2002). Consequently, in line with much of the related literature, we employ intraday changes in the 3-month swap rate as our external instrument. It is important to note, however, that the use of differing sample periods may raise concerns. Potential issues such as structural breaks, shifts in economic dynamics, and variations in policy regimes between the two periods should be considered. These caveats must be taken into account when interpreting our results.

[29] The choice of this period is also motivated by the structural changes observed in monetary policy transmission

estimates introduced by extreme observations in real variables during the COVID-19 period. In addition, we include one lag in the VAR model due to concerns about a weak instruments problem associated with considering a higher number of lags.[30] The estimated VAR model is stable as the inverse roots of the characteristic polynomial have modulus less than one and lie within the unit circle. Furthermore, the null hypothesis of no serial correlation of the residuals cannot be rejected according to the LM test.

3 Impulse Response Functions

3.1 The Effects of Monetary Policy in Mexico

In this section, we present the impulse responses of output, prices, the one-year interest rate, and the exchange rate to a restrictive monetary policy shock, identified using surprises in the 3-month swap rate as an external instrument in the VAR model. For comparison, we also report the estimated responses of these variables to a contractionary monetary policy shock identified using the standard Cholesky identification scheme, which has been a widely used approach in VAR studies of Mexican monetary policy. In the Cholesky scheme, the one-year interest rate is ordered after output and prices, while the exchange rate follows subsequently.

following the implementation of an inflation targeting regime by Banco de México starting in 2001 (Gaytán and González García, 2007). The pursuit of low and stable inflation has instilled confidence in financial contracts, thereby reducing the risk premium associated with interest rates and allowing for longer term contracts. Additionally, the central bank's inflation target of 3% was officially established in 2003 and has since served as a guiding principle for monetary policy decisions.

[30] In determining the optimal lag length for our VAR model, we initially consider the Bayesian information criterion (BIC) and the Hannan-Quinn criterion, both of which suggest a lag length of two. However, opting for two lags raises concerns about a weak instruments problem associated with our instrument, as indicated by the F-statistic falling below the threshold value of 10 suggested by Stock et al. (2002). Given this concern, we choose to include only one lag in the VAR model. This choice also reflects considerations regarding the estimation method and the inference procedure. Nonetheless, we confirm that our main results are robust when estimating the model with two lags. The results from this alternative specification are reported in Figures B.9 to B.10 of Appendix B. Another alternative to address concerns about the role of the VAR structure would be to compare our VAR-based impulse responses with those obtained from local projections. However, unlike the VAR framework which allows the model to be estimated over the full 2004–2019 sample even though the instrument is available only from 2011–2019, local projections must be estimated over the effective sample in which the instrument is observed, thereby restricting the estimation to the shorter 2011–2019 window. As a consequence, the resulting impulse responses would be subject to greater estimation uncertainty. For these reasons, and because the results would not be fully comparable due to differences in the estimation sample, we do not pursue this exercise.

This recursive structure implies that economic activity and inflation respond with a lag to the monetary policy shock, and that the central bank considers the current stage of output and prices in its decision-making. In turn, the adjustment of the one-year interest rate can immediately impact the exchange rate. However, contemporaneous innovations in the exchange rate do not affect how the central bank influences the monetary policy indicator.[31] In both cases—the external instruments approach and the Cholesky identification scheme—the magnitude of the monetary policy shock is standardized to one percentage point increase in the one-year interest rate.[32] We present these responses along with 68 percent confidence intervals, estimated using bootstrapping methods.[33]

Panel A of Figure 1 illustrates the responses of the aforementioned variables to a contractionary monetary policy shock, identified using the external instruments approach. As previously indicated, we employ intraday changes in the 3-month swap rate as our external instrument to identify the monetary policy shock. This instrument explains nearly 9% of the monthly variation in the reduced-form residuals for the one-year interest rate and has an associated robust F-statistic of 12.27.[34] This value is above the threshold of 10 suggested by Stock et al. (2002), thereby mitigating concerns regarding weak instrument problems in our

[31] This ordering assumption is recurrent in the literature, grounded in the principle that economic variables, such as output and prices, take time to adjust to monetary policy shocks (Friedman, 1961). On the other hand, financial variables like the exchange rate react quickly to changes in monetary policy in economies with a floating regime (Carrillo and Elizondo, 2015).

[32] Standardizing the size of the monetary policy shock serves several purposes. Firstly, it allows for a direct comparison between both approaches. Secondly, it facilitates the comparability of our findings with those in the literature, where a one percentage point increase is commonly used as the standard. Thirdly, it enables us to assess the magnitude of the effects of monetary policy surprises on macroeconomic variables. As commonly observed in the HFI literature, our monetary policy surprises are relatively small compared to raw changes in interest rates, suggesting that market participants anticipate the majority of policy rate decisions (Nakamura and Steinsson, 2018). Specifically, within the external instruments approach, a one-standard-deviation monetary policy shock is associated with a 0.05 percentage point increase in the 3-month swap rate and a 0.12 percentage point increase in the one-year interest rate. Conversely, in the Cholesky identification scheme, a one-standard-deviation monetary policy shock corresponds to a 0.19 percentage point increase in the one-year interest rate. In Figure B.11 of Appendix B, we present the estimated impulse responses without standardization.

[33] We follow the approach taken by Jentsh and Lunsford, 2019, employing moving block bootstrap. The estimation errors associated with the instrumental variable regression are encompassed in the reported confidence intervals, as both stages of the estimation process are included in the bootstrapping procedure. We employ 1,000 bootstrap replications to estimate the confidence intervals.

[34] The explanatory power of this instrument is similar to that used by Gertler and Karadi (2015) for the US. In particular, they find that the three-month-ahead futures rate surprises explain around 8% of the variation in the one-year interest rate.

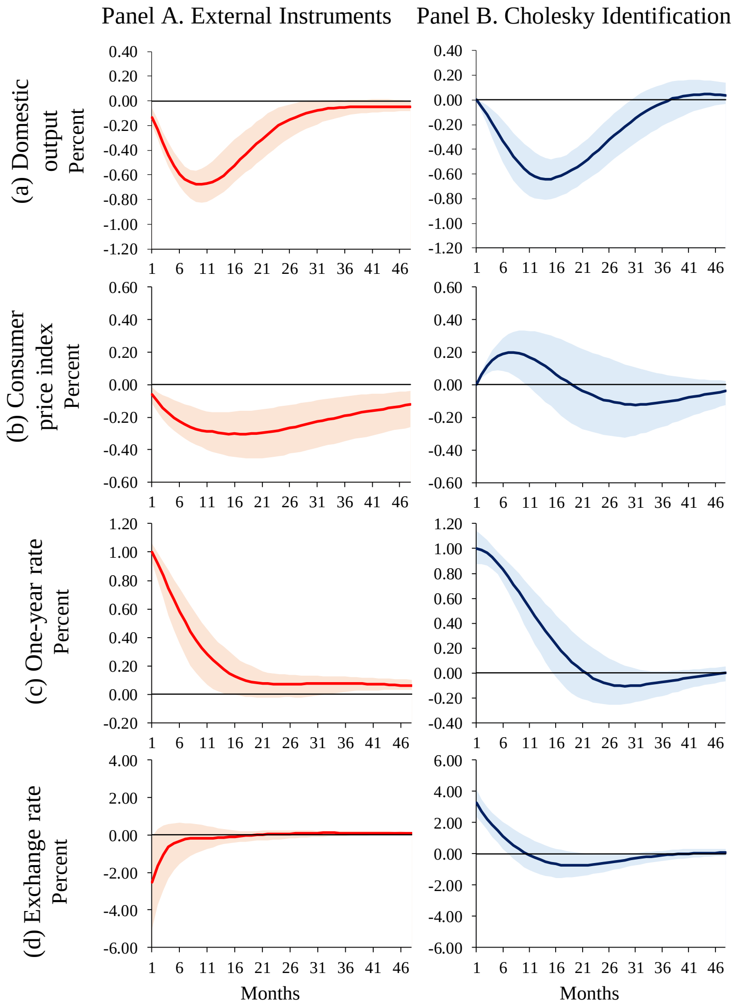
analysis. As depicted in Panel A of Figure 1(a), Mexican real activity undergoes a reduction following a restrictive monetary policy shock. Specifically, real output exhibits a U-shaped response and reaches its lowest point in the tenth month, registering a decline ranging from 0.55% to 0.83% below the baseline. In turn, the response of this variable is statistically distinguishable from zero on impact and appears to gradually converge to the baseline about two and a half years after the shock.[35] This result is in line with the standard transmission of monetary policy and is consistent with Nguyen (2020), who reports an average maximum output drop in EMEs of approximately 0.82% after a one-percentage-point increase in the monetary policy rate, typically observed around 9 months post-shock.

Consistent with conventional economic theory, prices also decrease in response to a restrictive monetary policy shock, as shown in Panel A of Figure 1(b). The maximum decline in this variable is observed 18 months after the shock, ranging from 0.16% to 0.45% below the baseline. The response of prices becomes statistically distinguishable from zero on impact and gradually converges to the baseline. This finding aligns with Havranek and Rusnak (2018), who analyze VAR results on the timing of price declines following a monetary policy shock across emerging and advanced economies. Specifically, they report an average transmission lag of around 18 months for hump-shaped impulse responses that do not exhibit a "price puzzle".[36]

Regarding the response of the exchange rate to a contractionary monetary policy shock,

[35] While the impact response is statistically different from zero, its magnitude is relatively modest, with a median response of about -0.13%. This finding is consistent with Capistrán et al. (2019), who also document a statistically significant impact response of output to a monetary policy shock in Mexico using a structural vector error correction model. More broadly, our findings also align with the view that the effects of monetary policy on real activity reach their peak with a lag, whereas impact effects are generally small, as documented, for instance, by Gertler and Karadi (2015).

[36] Compared to the findings of Deb et al. (2023) for a broad sample of advanced and emerging economies, our estimated inflation response is of similar magnitude, as their estimates range from 0.2 to 0.4 percent. In contrast, the output effects observed for Mexico are larger, since Deb et al. (2023) report output reductions within the same 0.2 to 0.4 percent range. This difference may reflect how the monetary policy shock is standardized. In our case, it corresponds to a one percentage point increase in the one-year interest rate, whereas Deb et al. (2023) define it as a one percentage point increase in short-term interest rates. More broadly, studies employing high-frequency identification strategies that standardize the shock as a one percentage point increase in the one-year rate tend to find stronger monetary policy effects than those based on traditional approaches. For instance, Checo et al. (2024), using data from both advanced and emerging economies, report output responses between 0.5 and 4 percent, and inflation responses ranging from 2 to 4 percent.



First-stage regression: F: 10.78; Robust F: 12.27; R^2: 9.39%; $\bar{R}^2$: 8.52%

Figure 1: Impulse Response Functions to a Monetary Policy Shock

Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.

the Mexican peso immediately appreciates against the US dollar with confidence intervals ranging from -0.01% to -5.02%, and a median response of -2.53%, followed by a gradual depreciation returning to the original level (see Panel A of Figure 1(d)).[37] This estimated

[37] Notice that, although this response is statistically significant at the 68% confidence level, it is subject to estimation uncertainty. This is consistent with Solís (2023a), who shows for Mexico that the exchange rate response to monetary policy shocks is subject to high estimation uncertainty when low frequency exchange rate data is used, and is more precisely identified using intraday data, using event study regressions. Therefore, the use of monthly exchange rate data, as required within the VAR framework with macroeconomic variables, may contribute to the relatively high variance of the estimated response of the exchange rate.

pattern of exchange rate movement aligns closely with the findings of Bjørnland (2009), Capistrán et al., (2019), and Ha and So (2020), thus supporting the overshooting theory initially proposed by Dornbusch (1976). Moreover, the strong appreciation of the Mexican currency in response to a contractionary monetary policy shock is also consistent with Solís (2023a, 2023b).[38] This result underscores the potential role of the exchange rate as a transmission channel for monetary policy in Mexico, an aspect we explore further in the next subsection. [39] It is worth mentioning that, unlike the Cholesky identification, we do not impose zero restrictions on the contemporaneous responses of output and prices to a monetary policy shock when employing the external instruments approach. The use of this alternative method to identify the aforementioned shock is crucial for obtaining these results.

Panel B of Figure 1 presents the responses to a contractionary monetary policy shock identified under the Cholesky scheme. In this identification scheme, the one-year interest rate is ordered after output and prices, and the exchange rate is placed after the interest rate. As illustrated in this figure, the identification of a monetary policy shock under the Cholesky scheme using only the variables considered thus far seems to be questionable. Notably, prices and the exchange rate present a puzzle; the monetary policy shock is followed by a statistically significant increase in prices and a persistent depreciation of the Mexican peso.[40] Although the point estimate of prices eventually declines, the maximum decrease in this variable occurs two and a half years after the shock, with a median response of about -0.12% with confidence intervals that include zero, ranging from -0.31% to 0.10%. This

[38] When rescaled to a one-standard-deviation monetary policy shock, the estimated contemporaneous exchange rate response corresponds to a median appreciation of 0.32%. This estimate is consistent with the distribution of the absolute daily exchange rate changes observed on monetary policy announcement days in Mexico over the 2004–2019 sample, for which the median change is 0.34% and the interquartile range spans from 0.16% to 0.56%.

[39] According to Boivin et al. (2010), the sensitivity of the exchange rate to interest rate movements is a crucial factor in understanding the exchange rate's role as a transmission channel of monetary policy. The authors suggest that when the estimated sensitivities are large, the transmission channel is important, and vice versa.

[40] A common methodological approach proposed in the literature to mitigate the price puzzle frequently observed in VAR models identified through recursive schemes is to incorporate forward-looking variables, such as inflation expectations, commodity price, or intermediate-goods price measures, to better capture anticipatory monetary policy responses (see Sims, 1992; Sims and Zha, 2006; Christiano et al., 2005; Canova and De Nicolo, 2002). For the case of Mexico, Carrillo and Elizondo (2015) employ producer prices. To evaluate the robustness of the external instruments approach in models with richer information sets, we include the producer price index as an additional endogenous variable. The results of this exercise are presented in Section 3.3.

indicates a considerably slower and weaker transmission of monetary policy, which appears inconsistent with the findings of the cross-country survey study reported by Havranek and Rusnak (2018).

Regarding the behavior of the exchange rate, the persistent depreciation of the Mexican peso after the monetary policy shock seems inconsistent with Dornbusch (1976) overshooting hypothesis. Given the comprehensive information integrated into our VAR model, it seems more plausible that there is an issue with the identifying restrictions, as in Gertler and Karadi (2015).[41] It appears that the central bank is responding, at least partially, to the information contained in the exchange rate. A depreciation of the Mexican peso puts upward pressure on prices. If significant, the central bank may tighten to avoid an increase in inflation. The resulting price puzzle is consistent with this interpretation. Consequently, the restriction that the central bank does not respond to innovations in the exchange rate does not seem reasonable. Conversely, it would not be sensible to assume that the exchange rate cannot respond instantly to a monetary policy shock, as implied by the efficient markets hypothesis. Thus, in line with Gertler and Karadi (2015), our results also indicate that the timing restrictions used in VAR models for EMEs may not work well in scenarios where both macroeconomic and financial variables, like the exchange rate, are involved. In contrast, the external instruments approach appears well-suited for this type of environment.

3.2 The Exchange Rate Channel as a Transmission Mechanism of Monetary Policy in Mexico

The previous findings provide valuable insights into the timing and effects of monetary policy on the Mexican economy. In this subsection, we evaluate the importance of the exchange

[41] To assess the usefulness of the external instruments approach, even when the VAR model incorporates a more comprehensive set of information, we conduct two additional robustness checks in Section 3.4. Specifically, we separately include an indicator of US financial conditions and a measure of commodity prices. The inclusion of the first variable helps account for shifts in US financial conditions, which, due to the strong economic ties between Mexico and the US, can significantly influence local markets and shape the monetary policy response in Mexico. On the other hand, adding an indicator of commodity prices allows us to capture future inflation trends, enriching the information set available to the central bank and potentially addressing the price puzzle commonly found in the VAR literature. Our results remain consistent with the baseline specification.

rate channel as a transmission mechanism of monetary policy. As mentioned earlier, this channel is particularly relevant for small open economies like Mexico, which heavily rely on international trade. Adjustments in the exchange rate can significantly impact import prices, export competitiveness, and consequently, overall economic conditions such as economic activity and inflation. Thus, understanding how the central bank influences the economy through this channel may provide valuable insights for the design of monetary policy.

According to the exchange rate channel, when the central bank increases the monetary policy rate, domestic financial assets become relatively more attractive than foreign ones. As a result, the domestic currency tends to appreciate as the demand for domestic assets increases relative to foreign assets. The appreciation of the domestic currency can, in turn, reduce domestic prices through various mechanisms.[42] The first one operates directly through the decrease in the prices of imported goods and inputs used in domestic consumption and production. When the local currency appreciates, imported goods become cheaper, leading to lower prices for these goods in the domestic market. This may lead to a moderation in inflation. The second mechanism operates indirectly through aggregate demand. The lower price of the foreign currency discourages exports and stimulates imports.[43] As a result, the demand for domestic goods falls, reducing the pressure on the price level. Finally, another mechanism by which movements in the exchange rate may be transmitted to inflation works indirectly through their effect on inflation expectations (Sidaoui and Ramos-Francia, 2008). Specifically, an appreciation of the domestic currency could lead to a downward revision in short-term inflation expectations, thereby exerting downward pressure on inflationary pressures. In general, the strength of the exchange rate channel depends on several factors, including the responsiveness of the exchange rate to monetary policy, the degree of openness of the economy, the exchange rate pass-through to prices, and the sensitivity of net exports to

[42] Mishkin (1995), Taylor (1995), and more recently Boivin et al. (2010) present a detailed discussion of several of these mechanisms.

[43] This mechanism is consistent with standard small open economy frameworks, where exchange rate movements affect net exports through relative prices and competitiveness effects. In particular, a depreciation tends to increase net exports by making domestic goods cheaper for foreign buyers and imports more expensive for residents, whereas an appreciation has the opposite effect (Galí and Monacelli, 2005).

exchange rate variations.[44]

To examine the importance of the exchange rate channel in the transmission of monetary policy within the Mexican economy, we adopt an exogenization procedure that turns off this channel. In particular, from Eq. (3), we treat the exchange rate as exogenous, including its lags as predetermined regressors in the remaining endogenous equations. In this way, the procedure blocks both the contemporaneous effect of the identified monetary policy shock on the exchange rate and any subsequent transmission operating through the endogenous interaction of other variables with the exchange rate.[45] This yields a system whose estimated parameters coincide with those in the unrestricted model, which makes the two specifications directly comparable (see Appendix C). We then estimate impulse response functions for the restricted model using the same external instrument approach. Importantly, both models are characterized by similar orthogonalized innovations, given by surprises in the 3-month swap rate. Accordingly, the impulse responses in the restricted model may differ from those of the unrestricted one only because any transmission running through the endogenous response of the exchange rate is now blocked. Hence, comparing the two sets of impulse responses provides a measure of the importance of the exchange rate channel. Note that, this exercise may be susceptible to the Lucas critique, and that is intended to capture the marginal contribution of the exchange rate to the propagation of the monetary policy shock.

Furthermore, to assess whether the differences between impulse responses implied by the

[44] Several empirical studies for Mexico, including Sidaoui and Ramos-Francia (2008), Capistrán et al. (2012), and Cortés (2013), provide evidence that the exchange rate pass-through to consumer prices has remained low since the adoption of an inflation-targeting regime in 2001. This reduced pass-through is commonly attributed to the increased credibility of monetary policy, which has contributed to well-anchored inflation expectations. In such an environment, firms are less inclined to adjust prices following currency depreciations, as they anticipate that the central bank will act firmly to preserve price stability.

[45] Related contributions, such as Morsink and Bayoumi (2001), Disyatat and Vongsinsirikul (2003), Aleem (2010), and Ibarra (2016), employ similar exogenization schemes to assess specific monetary policy transmission channels in Japan, Thailand, India, and Mexico, respectively. In our setting, exogenizing the exchange rate by suppressing its equation and treating its lags as given in the remaining endogenous equations blocks both the monetary policy shock from affecting the exchange rate on impact and the lagged feedback from the endogenous system into it. This approach is related to that employed by Ramey (1993), Uribe and Yue (2006), Degasperi et al. (2023), among others, with the key difference that their "zeroing-out" methodology sets to zero only the coefficients that map a given structural shock into the variable of interest. This shuts down the impact response of that variable to the shock but does not, in general, eliminate all lagged transmission operating through the system.

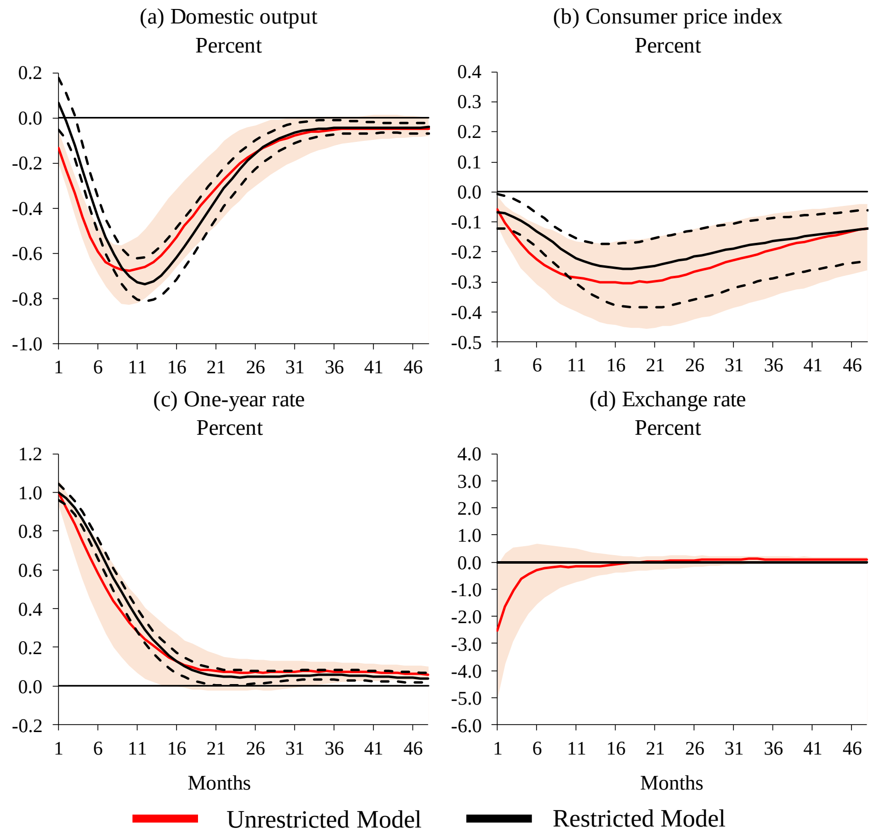
unrestricted and restricted models are statistically significant, we implement a Wald test with a bootstrap covariance matrix, following Trenkler and Weber (2020) and Jordà (2009). Let $\hat{\theta}_{m,y}(h)$ denote the impulse response of variable y to a monetary policy shock at horizon h in model $m \in \{1,2\}$, where $m = 1$ corresponds to the unrestricted model and $m = 2$ to the restricted one. We test the null hypothesis that the impulse responses in the two models are equal at all horizons in $H = \{1,\ldots,48\}$, that is,

$$H_0: \ \theta_{1,y}(h) - \theta_{2,y}(h) = 0 \quad \forall h \in H. \tag{7}$$

Let $\hat{\delta}_y(H)$ denote the vector that stacks the differences between the point estimates of the impulse responses across models, and let $\hat{S}_y(H)$ be its covariance matrix, computed from the bootstrap replications. The Wald statistic is then $W_y(H) = \hat{\delta}_y(H)'\hat{S}_y(H)^{-1}\hat{\delta}_y(H)$. To obtain the corresponding bootstrap Wald statistics, for each replication b we center the bootstrap differences $\hat{\delta}_y^{(b)}(H)$ around their mean and compute $W_y^{(b)}(H)$ in the same way. Accordingly, the p-value is given by the proportion of $W_y^{(b)}(H)$ that exceed $W_y(H)$.

Figure 2 illustrates the responses of the aforementioned variables to a restrictive monetary policy shock in these two models: the unrestricted (red lines) and the restricted models (black lines). As before, the magnitude of the monetary policy shock is standardized to one percentage point increase in the one-year interest rate. It is worth noting that, by construction, the F-statistics and the R-squared values are identical in both models. Additionally, in the restricted model, the response of the exchange rate is null, as depicted in Figure 2(d). Notably, the effects of a restrictive monetary policy shock on prices appear considerably weaker and more delayed when the exchange rate channel is turned off (see Figure 2(b)). Specifically, relative to the unrestricted model, the median response of prices is almost 50% smaller in the sixth month, and around 16% smaller at its point of maximum decline, in month eighteen, where it ranges from 0.17% to 0.35% below the baseline. These results suggest that by allowing the exchange rate channel to operate appears to result in a notably faster and stronger price response to a contractionary monetary policy shock.[46]

[46] When the exchange rate channel is turned off, the median effect of the monetary policy shock on prices approx-



First-stage regression: F: 10.78; Robust F: 12.27; R^2: 9.39%; $\bar{R}^2$: 8.52%

Figure 2: Impulse Response Functions to a Monetary Policy Shock estimated using the Unrestricted and the Restricted Model with the Exogenized Exchange Rate

Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics and the R-squared values correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.

Regarding the response of real output to a contractionary monetary policy shock, in the unrestricted model, this variable reacts more rapidly to the aforementioned shock. As observed in Figure 2(a), the response of real output becomes statistically distinguishable from zero on impact—three months earlier than in the restricted model. After three months, when the exchange rate channel is operative, real output exhibits a decline ranging between 0.24%

imately one year later is about 0.064 percentage points smaller than in the unrestricted model. Given that the initial median impact of the shock on the exchange rate is 2.5%, this implies a pass-through elasticity of about 0.03. This estimate aligns with previous findings by Cortés (2013), Kochen and Sámano (2016), Angeles et al. (2019), and Jaramillo et al. (2019), who report pass-through elasticities in the range of 0.03 to 0.06.

and 0.42% below the baseline, a substantial deviation that likely contributes to the stronger decrease in prices observed in the unrestricted model. This result is expected, given that exchange rate dynamics can indirectly impact aggregate demand. The strong appreciation of the Mexican peso in response to a contractionary monetary policy shock discourages exports and stimulates imports, consequently diminishing the demand for domestic goods and reducing the price level.

Moreover, consistent with the visual differences, the Wald test for equality of impulse responses over all the horizon applied to the one-year interest rate, real output, and prices yields p-values of 0.01, 0.00, and 0.00, respectively, while a joint test across all variables yields a p-value of 0.00. Hence, we reject the null that the two sets of impulse responses are identical at conventional significance levels, indicating that turning off the exchange rate channel may lead to statistically significant changes in the dynamic effects of a monetary policy shock.

Overall, our results are consistent with Capistrán et al. (2019) and may suggest that an important mechanism through which the exchange rate influences inflation in Mexico could be fundamentally linked to its direct effects on the price level or on inflation expectations. Note, however, that in the restricted VAR model, while real output seems to react more slowly in response to a restrictive monetary policy shock, the maximum decrease in this variable is approximately between 0.62% and 0.81% below the baseline after twelve months, which is similar to the maximum decrease observed in the benchmark (0.53% to 0.83%, in the tenth month). Thus, it is likely that the appreciation of the exchange rate influences inflation through channels other than net exports or aggregate demand. Specifically, the strong appreciation of the Mexican peso in response to a contractionary monetary policy shock may lead to a decrease in the prices of imported goods, contributing to a moderation in inflationary pressures.[47] Furthermore, the appreciation of the domestic currency could prompt a down-

[47] Based on our calculations using World Development Indicators data for 2004–2019, Mexico's imports of goods and services average about 32.6% of GDP, and imports of goods about 29.3% of GDP. For comparison, Mexico's imports of goods as a share of GDP rank among the top 6 of the 24 emerging economies included in the MSCI Emerging Markets Index, and are the highest among the Latin American economies in that group. This relative position suggests that Mexico displays a relatively high degree of trade openness compared with other emerging market economies. In an economy with this degree of openness, a sizable share of household consumption and

ward adjustment in inflation expectations, thereby exerting further downward pressure on inflation. This suggests a complex interplay between exchange rate dynamics, inflation, and monetary policy, highlighting the importance of considering multiple mechanisms through which exchange rate movements can impact the macroeconomic environment.

3.3 Additional Robustness Exercises

In this subsection, we introduce some additional robustness checks. Specifically, we examine the sensitivity of our main findings to the inclusion of an additional indicator of foreign financial conditions. Additionally, we assess the usefulness of the external instruments approach, even when the VAR model includes a more comprehensive set of information. This involves the inclusion of a measure of prices for intermediate goods as an additional endogenous variable. Finally, we analyze the impact of central bank information regarding the economic outlook on the effects of monetary policy in Mexico.

3.3.1 Additional Indicator of Foreign Financial Conditions

In this exercise, we introduce an additional exogenous variable: the US excess bond premium (EBP), as introduced by Gilchrist and Zakrajšek (2012). This premium represents the average corporate bond spread not attributable to default risk. According to Gilchrist and Zakrajšek (2012) and Favara et al. (2016), it provides an effective measure of investor sentiment or risk appetite in the US corporate bond market. Furthermore, this variable aggregates high-quality forward-looking information about the US economy (Gilchrist and Zakrajšek, 2012; Favara et al., 2016), enhancing the reliability and forecasting performance of small-scale VARs (Caldara and Herbst, 2019). Given the strong financial integration between Mexico and the US, including the excess bond premium in the VAR model is particularly relevant. This approach allows us to account for exogenous shifts in financial conditions that may influence the design of monetary policy in Mexico.

production costs depends on imported goods, so changes in the exchange rate that alter the price of those imports are likely to pass through to domestic prices. This pattern is consistent with our interpretation that the exchange rate in Mexico mainly affects inflation via its impact on the prices of imported goods.

Overall, the results presented in Figures B.12 and B.13 of Appendix B align with our baseline specification. Our findings indicate that monetary policy shocks, identified using high-frequency surprises around central bank announcements in Mexico, lead to responses in output, prices and the exchange rate that align closely with conventional economic theory.[48] Importantly, we show that by employing the external instruments identification strategy, both the so-called price and exchange rate puzzles disappear. Additionally, our results suggest that the exchange rate channel seems to play an important role in the transmission of monetary policy in Mexico.

3.3.2 Additional Measure of Prices for Intermediate Goods

One common methodological solution proposed in the literature to mitigate the price puzzle frequently encountered in VAR models identified using a recursive scheme, involves incorporating a measure of prices for intermediate goods to capture future movements of inflation (see Sims and Zha, 2006; Christiano et al., 2005; Carrillo and Elizondo, 2015).[49] To assess the usefulness of the external instruments approach, even in scenarios where a VAR model incorporates a richer set of information, we include the producer price index for the Mexican economy as an additional endogenous variable in equation Eq. (3). Specifically, we employ the National Index for Producer Prices including oil obtained from INEGI.[50] Subsequently, we estimate the impulse responses of output, prices, and the exchange rate to a restrictive monetary policy shock, identified using both the external instruments approach and the Cholesky identification scheme. In the recursive scheme, all variables maintain their previous ordering, with the only difference being that producer prices are positioned after output and before consumer prices.

[48] Note, however, that when the EBP is included in the model, the effect of the monetary policy shock on output appears to be smaller than in the baseline specification. This pattern may reflect the role of the EBP in absorbing part of a possible US financial tightening that would otherwise be attributed to the domestic policy shock and mechanically amplify its estimated effect on activity.

[49] According to Sims (1992), the price puzzle may arise because central banks are forward-looking and react to anticipated future movements of inflation by adjusting the monetary policy rate. Thus, the omission of such inflation information in the VAR model results in monetary policy shocks that are not entirely exogenous, potentially leading to an increase in the interest rate followed by a rise in the price level.

[50] INEGI is the National Institute of Statistics and Geography in Mexico, which is the main agency responsible for collecting and disseminating statistical and geographical information in the country.

As shown in Figures B.14 and B.15 of Appendix B, the results align with our baseline specification.[51] Under the Cholesky identification scheme, while the inclusion of the producer price index in the model aids in mitigating the price puzzle, the response of prices to a contractionary monetary policy shock remains statistically insignificant over the entire horizon. Moreover, regarding the exchange rate response to a contractionary monetary policy shock, the inclusion of producer prices does not appear to alleviate the exchange rate puzzle. Hence, we conclude that the external instruments approach remains well-suited for this environment with VAR models incorporating more information.

3.3.3 Accounting for Central Bank Information Effects

An important finding in the recent literature is that monetary policy surprises are influenced not only by changes in the monetary policy stance, but also by central bank information about the economic outlook. By providing this additional information alongside the typical monetary policy stance, monetary authorities may further influence private expectations regarding both the economy and the future path of interest rates, thereby shaping investment decisions in the FX market and further impacting exchange rate dynamics. As illustrated in the previous section, through the exchange rate channel, monetary authorities in EMEs may exert significant influence on overall economic conditions such as economic activity and inflation.

In this section, we analyze the impact of central bank information regarding the economic outlook on the transmission of monetary policy in Mexico. Following Jarociński and Karadi (2020), we utilize the response of the Stock Market Index surrounding central bank announcements to identify monetary policy surprises that are purged from information effects. Intuitively, if the stock market index and interest rates move in the same direction, it may imply that the announcement was likely accompanied by information regarding the economic outlook. If the change is positive, it could signal that the central bank has tightened monetary policy and additionally signaled positive news about the economic outlook. As emphasized by Kerssenfischer (2022), in such cases, the negative discount rate effect on

[51] Although the F-statistic values below 10 may suggest a potential weak instrument issue, the robustness of the results offers reassurance that the impulse responses in this exercise are not spurious due to weak instruments.

stock prices is outweighed by the positive cash flow effect.[52] To implement this, we only use the monetary policy surprises during months when the Mexican Stock Market Index surprise around the announcement displays an opposite sign to the 3-month swap rate surprise, as a broad range of models predicts when monetary authorities unexpectedly adjust the policy rate. In turn, we assign a value of zero to the monetary policy surprises during months when both surprises in the 3-month swap rate and the stock price exhibit the same sign.[53,54] Once these purged surprises have been identified, we employ our VAR model to analyze their dynamic effects on output, prices, and the exchange rate.

The results presented in Figures B.16 and B.17 of Appendix B are consistent with our baseline specification, suggesting that our identification is not contaminated by central bank information effects. Our findings reveal that purged monetary policy shocks also lead to responses in output, prices, and the exchange rate that closely align with conventional macroe-

[52] Consider an illustrative example. On August 15, 2019, Banco de México surprised the market with a larger than expected 25-basis-point monetary policy rate cut. However, contrary to conventional expectations where such a move would lead to an upward move in stock prices, the Mexican Stock Market Index showed a decline within a 24-hour window, spanning from one day before to the same day of the announcement. Such an event is not unique: approximately 40% of the announcements from 2011 to 2019 exhibit this positive co-movement of interest rate and stock market changes. This observation becomes less surprising when we take into account the central bank's accompanying statement, which highlighted that "the negative output gap widened more than expected, and the balance of risks for economic growth continues to be skewed to the downside." In our interpretation, this pessimistic communication could lead to a fall in stock valuations in spite of the surprise policy easing.

[53] As described in Section 2.3, surprises in the 3-month swap rate are computed within a 30-minute window around the announcements. However, surprises in the Mexican Stock Market Index are calculated within a daily window due to the unavailability of intraday data. Specifically, these surprises are computed as the variation in the end-of-day value in the Stock Market Index, spanning from one day before to the same day of the policy announcement. Data are provided by the Mexican Stock Exchange Group (BMV, by its acronym in Spanish).

[54] This strategy aligns with recent studies highlighting the signaling channel of monetary policy (Cesa-Bianchi and Sokol, 2022; Melosi, 2017). An alternative approach followed by Jarociński and Karadi (2020) and Andrade and Ferroni (2021), for instance, consists of distinguishing between purged monetary policy shocks and concurrent central bank information shocks by imposing sign restrictions on high-frequency surprise data of interest rates and stock prices in a narrow window around central bank announcements. As a first approximation, however, we employ what Jarociński and Karadi (2020) and Ha and So (2023), among others, call the "poor man's" strategy as described before. Although the assumptions underlying this approach are stronger compared to those of sign restrictions, it is simpler to execute. Essentially, this strategy operates under the implicit assumption that each month is distinctly affected by either a purged monetary policy shock or a central bank information shock. In contrast, the sign restrictions approach allows for the potential coexistence of both types of shocks within each month. Note that, in our application, this procedure implies that only 40% of the original announcement observations remain. The resulting smaller sample, together with the combination of daily stock market surprises and intraday interest rate surprises, may introduce some measurement noise into the identification.

conomic models. Notably, by employing monetary policy surprises purged from information effects, both the price puzzle and exchange rate puzzle disappear. Furthermore, our analysis suggests that the exchange rate channel may continue to play an important role in the transmission of monetary policy in Mexico.[55]

4 Conclusion

In this article, we explore the transmission of monetary policy in Mexico. The analysis is conducted by estimating a VAR model that identifies monetary policy surprises around central bank decisions in Mexico as an external instrument to identify monetary policy shocks. Through this model, we derive the dynamic responses of output, prices, and the exchange rate to a contractionary monetary policy shock. Unlike the recursive identification scheme that has been commonly used in VAR studies of Mexican monetary policy, this approach does not require imposing timing restrictions in the VAR model to identify the monetary policy shock. This is particularly crucial in the context of EMEs, where imposing timing restrictions has often resulted in puzzling outcomes that are difficult to reconcile with conventional economic theory. As a by-product of our analysis, we also analyze the transmission of monetary policy to the Mexican economy through the exchange rate channel. This channel holds particular significance for open economies like Mexico, which heavily rely on international trade.

The main results indicate that monetary policy shocks, identified using high-frequency surprises around central bank decisions in Mexico, lead to responses in output, prices, and the exchange rate that align closely with conventional macro models. Importantly, we show that by employing the external instruments identification strategy, both the so-called price and

[55] Moreover, to assess the role of the exchange rate in the transmission of monetary policy in the specification purged from central bank information effects, we implement the Wald test for equality of impulse responses between the restricted model that blocks off the effects of monetary policy on the exchange rate and the unrestricted model that allows for these effects, following the methodology described in Section 3.2. The resulting Wald tests over all horizons applied to the one-year interest rate, real output, and prices yield p-values of 0.00, 0.01, and 0.01, respectively, while a joint test across all variables yields a p-value of 0.00. These results lead us to reject the null of equal impulse responses at conventional significance levels, indicating that shutting down the exchange rate channel may lead to statistically significant differences in the effects of a monetary policy shock purged of central bank information effects on macroeconomic variables.

exchange rate puzzles disappear, consistent with the literature on monetary policy transmission that has documented alternative ways to mitigate these puzzles. Following a restrictive monetary policy shock, Mexican output and inflation display a U-shaped response, reaching their maximum reduction after a lag. Regarding the response of the exchange rate to a contractionary monetary policy shock, the Mexican peso immediately appreciates against the US dollar, followed by a gradual depreciation until it returns to the original level. These results are in line with the standard transmission of monetary policy.

Regarding the importance of the exchange rate channel as a transmission mechanism of monetary policy in Mexico, our findings suggest that this channel may strengthen the response of prices to a contractionary monetary policy shock, underscoring its relevance for the conduct of monetary policy in a small open economy such as Mexico. These results are consistent with Capistrán et al. (2019) and may suggest that an important mechanism through which the exchange rate influences inflation in Mexico is linked to its direct effects on prices and inflation expectations.

Our findings align with those of Gertler and Karadi (2015), indicating that the external instrument approach to analyze the effects of monetary policy could help address the price and exchange rate puzzles frequently encountered in the VAR literature for EMEs. Further research could explore the transmission of monetary policy in other EMEs using the external instruments approach as financial markets develop and data availability improves. Additionally, investigating alternative transmission channels could yield valuable insights. It would also be pertinent to analyze how the transmission of monetary policy has evolved in the aftermath of the COVID-19 pandemic. Furthermore, examining the effects of monetary policy shocks on additional macroeconomic indicators would contribute to a more comprehensive understanding of monetary policy dynamics. Lastly, conducting an analysis of the effects of monetary policy shocks and information shocks regarding the economic outlook in EMEs using alternative methodologies could help in enhancing our understanding of monetary policy transmission in these economies.

References

[1] Aleem, A. (2010). Transmission mechanism of monetary policy in India. *Journal of Asian Economics* **21**(2), 186-197.

[2] Andrade, P., and Ferroni, F. (2021). Delphic and odyssean monetary policy shocks: Evidence from the euro area. *Journal of Monetary Economics* **117**, 816-832.

[3] Angeles Galván, D., Cortés Espada, J. F., and Sámano, D. (2019). Evolución y características del traspaso del tipo de cambio a precios en México. (No. 2019-10). Banco de México Working Paper.

[4] Andersen, T. G., Bollerslev, T., Diebold, F. X., and Vega, C. (2003). Micro effects of macro announcements: Real-time price discovery in foreign exchange. *American economic review* **93**(1), 38-62.

[5] Andersen, T. G., Bollerslev, T., Diebold, F. X., and Vega, C. (2007). Real-time price discovery in global stock, bond and foreign exchange markets. *Journal of international Economics* **73**(2), 251-277.

[6] Banco de México (2008). Annual inflation report.

[7] Banco de México (2021). Quarterly inflation report. April-June.

[8] Barakchian, S. M., and Crowe, C. (2013). Monetary policy matters: Evidence from new shocks data. *Journal of Monetary Economics* **60**(8), 950-966.

[9] Bauer, M. D., and Rudebusch, G. D. (2016). Monetary policy expectations at the zero lower bound. *Journal of Money, Credit and Banking*, **48**(7), 1439-1465.

[10] Bauer, M. D., and Swanson, E. T. (2023a). An alternative explanation for the "fed information effect". *American Economic Review* **113**(3), 664-700.

[11] Bauer, M. D., and Swanson, E. T. (2023b). A reassessment of monetary policy surprises and high-frequency identification. *NBER Macroeconomics Annual* **37**(1), 87-155.

[12] Bernanke, B. S., & Blinder, A. S. (1992). The Federal Funds Rate and the Channels of Monetary Transmission. *American Economic Review*, **82**(4), 901–921.

[13] Bernanke, B. S., & Gertler, M. (1995). Inside the Black Box: The Credit Channel of Monetary Policy Transmission. *Journal of Economic Perspectives*, **9**(4), 27–48.

[14] Bernanke, B. S., and Kuttner, K. N. (2005). What explains the stock market's reaction to Federal Reserve policy? *The Journal of Finance* **60**(3), 1221-1257.

[15] Bjørnland, H. C. (2009). Monetary policy and exchange rate overshooting: Dornbusch was right after all. *Journal of International Economics* **79**(1), 64-77.

[16] Blinder, A. S., Ehrmann, M., Fratzscher, M., De Haan, J., and Jansen, D. J. (2008). Central bank communication and monetary policy: A survey of theory and evidence. *Journal of Economic Literature* **46**(4), 910-945.

[17] Boivin, J., Kiley, M. T., and Mishkin, F. S. (2010). How has the monetary transmission mechanism evolved over time? *Handbook of monetary economics* **3**, 369-422.

[18] Bredin, D., Hyde, S., and Reilly, G. O. (2010). Monetary policy surprises and international bond markets. *Journal of International Money and Finance* **29**(6), 988-1002.

[19] Caldara, D., and Herbst, E. (2019). Monetary policy, real activity, and credit spreads: Evidence from Bayesian proxy SVARs. *American Economic Journal: Macroeconomics* **11**(1), 157-92.

[20] Campbell, J. R., Evans, C. L., Fisher, J. D., Justiniano, A., Calomiris, C. W., and Woodford, M. (2012). Macroeconomic effects of federal reserve forward guidance. *Brookings Papers on Economic Activity* **42**(1), 1-80.

[21] Canova, F., and De Nicolo, G. (2002). Monetary disturbances matter for business fluctuations in the G-7. *Journal of Monetary Economics* **49**(6), 49(6), 1131-1159.

[22] Capistrán, C., Ibarra, R., and Ramos Francia, M. (2012). El traspaso de movimientos del tipo de cambio a los precios. Un análisis para la economía mexicana. *El Trimestre Económico* **79**(316), 813-838.

[23] Capistrán, C., Chiquiar, D., and Hernández, J. R. (2019). Identifying Dornbusch's exchange rate overshooting with structural VECs: evidence from Mexico. *61st issue (December 2019) of the International Journal of Central Banking*.

[24] Carrillo, J. A., and Elizondo, R. (2015). How robust are SVARs at measuring monetary policy in small open economies? (No. 2015-18). Banco de México Working Paper.

[25] Carrillo, J. A., Elizondo, R., and Hernández-Román, L. G. (2020). Inquiry on the transmission of US aggregate shocks to Mexico: a SVAR approach. *Journal of International Money and Finance* **104**, 1-24.

[26] Carrillo, J., Elizondo, R., Hernández-Román, L., Ibarra, R., and Quiroga, M. (2025). Monetary policy transmission in a small open economy before and after the pandemic: the case of Mexico. Banco de México, mimeo.

[27] Cesa-Bianchi, A., Thwaites, G., and Vicondoa, A. (2020). Monetary policy transmission in the United Kingdom: A high frequency identification approach. *European Economic Review* **123**, 103375.

[28] Cesa-Bianchi, A., and Sokol, A. (2022). Financial shocks, credit spreads, and the international credit channel. *Journal of International Economics* **135**, 1-18.

[29] Céspedes, B., Lima, E., and Maka, A. (2008). Monetary policy, inflation and the level of economic activity in Brazil after the real plan: stylized facts from SVAR models. *Revista Brasileira de Economia* **62**, 123-160.

[30] Checo, A., Grigoli, F., and Sandri, D. (2024). Monetary policy transmission in emerging markets: proverbial concerns, novel evidence (No. 1170). Bank for International Settlements Working Papers.

[31] Chiguil-Rojas, A., Esquivel, G., and Leal, J. (2024). La transmisión de la política monetaria a través del crédito bancario en México. *El Trimestre Económico* **91**(363), 603-662.

[32] Christiano, L. J., Eichenbaum, M., and Evans, C. L. (2005). Nominal rigidities and the dynamic effects of a shock to monetary policy. *Journal of Political Economy* **113**(1), 1-45.

[33] Colunga, L. F. (2020). The loan puzzle in México. *Working Papers*.

[34] Copelman, M., and Werner, A. M. (1997). El mecanismo de la transmisión monetaria en México. *El Trimestre Económico* **64**(253), 75-104.

[35] Cortés, J. F. (2013). Estimación del traspaso del tipo de cambio a los precios en México. *Monetaria* **35**(2), 312-4344.

[36] Cuadra, G. (2008). Hechos estilizados del ciclo económico en México. *Banco de México, Documentos de Investigación* **2008-14**.

[37] Cushman, D. O., and Zha, T. (1997). Identifying monetary policy in a small open economy under flexible exchange rates. *Journal of Monetary Economics* **39**(3), 433-448.

[38] Deb, M. P., Estefania-Flores, J., Firat, M., Furceri, D., and Kothari, S. (2023). Monetary policy transmission heterogeneity: cross-country evidence (WP/23/204). International Monetary Fund Working Papers.

[39] Degasperi, R., Hong, S. S., and Ricco, G. (2023). The global transmission of U.S. monetary policy. CREST Working Paper 2023-02.

[40] Díaz de León, A., and Greenham, L. (2001). Política monetaria y tasas de interés: experiencia reciente para el caso de México. *Economía Mexicana Nueva Época* **10**(2), 213-258.

[41] Disyatat, P., and Vongsinsirikul, P. (2003). Monetary policy and the transmission mechanism in Thailand. *Journal of Asian Economics* **14**(3), 389-418.

[42] Dornbusch, R. (1976). Expectations and exchange rate dynamics. *Journal of Political Economy* **84**(6), 1161-1176.

[43] Ehrmann, M., and Fratzscher, M. (2003). Monetary policy announcements and money markets: A transatlantic perspective. *International Finance* **6**(3), 309-328.

[44] Ehrmann, M., and Fratzscher, M. (2005). Equal size, equal role? Interest rate interdependence between the euro area and the United States. *The Economic Journal* **115**(506), 928-948.

[45] Eichenbaum, M., and Evans, C. L. (1995). Some empirical evidence on the effects of shocks to monetary policy on exchange rates. *The Quarterly Journal of Economics* **110**(4), 975-1009.

[46] Elizondo, R. (2012). Estimaciones del PIB mensual basadas en el IGAE (No. 2012-11). Banco de México Working Paper.

[47] Faust, J., Rogers, J. H., Wang, S. Y. B., and Wright, J. H. (2007). The high-frequency response of exchange rates and interest rates to macroeconomic announcements. *Journal of Monetary Economics* **54**(4), 1051-1068.

[48] Favara, G., Gilchrist, S., Lewis, K. F., and Zakrajsek, E. (2016). Recession risk and the excess bond premium. (No. 2016-04-08). Board of Governors of the Federal Reserve System (US).

[49] Ferrari, M., Kearns, J., and Schrimpf, A. (2021). Monetary policy's rising FX impact in the era of ultra-low rates. *Journal of Banking & Finance* **129**, 1-13.

[50] Fink, F., and Schüler, Y. S. (2015). The transmission of US systemic financial stress: Evidence for emerging market economies. *Journal of International Money and Finance* **55**, 6-26.

[51] Friedman, M. (1961). The lag in effect of monetary policy. *Journal of Political Economy* **69**(5), 447-466.

[52] Galí, J., and Monacelli, T. (2005). Monetary policy and exchange rate volatility in a small open economy. *Review of Economic Studies* **72**(3), 707–734.

[53] Garcés, D. (2001). Determinación del nivel de precios y la dinámica inflacionaria en México. *Monetaria* **24**(3), 241-269.

[54] Gaytán, A., and González García, J. R. (2007). Cambios estructurales en el mecanismo de transmisión de la política monetaria en México: un enfoque VAR no lineal. *Monetaria* **30**(4), 367-404.

[55] Gaytán, A., and González-García, J. (2007). Cambios estructurales en el mecanismo de transmisión de la política monetaria en México: Un enfoque VAR no lineal. *Monetaria* **30**(4), 367-404.

[56] Gertler, M., and Karadi, P. (2015). Monetary policy surprises, credit costs, and economic activity. *American Economic Journal: Macroeconomics* **7**(1), 44-76.

[57] Gilchrist, S., and Zakrajšek, E. (2012). Credit spreads and business cycle fluctuations. *American Economic Review* **102**(4), 1692-1720.

[58] Grilli, V., and Roubini, N. (1995). Liquidity and exchange rates: puzzling evidence from the G-7 countries (No. 95-17). Stern School of Business Working Paper.

[59] Gürkaynak, R. S., Sack, B., and Swanson, E. T. (2005). Do actions speak louder than words? The response of asset prices to monetary policy actions and statements. *International Journal of Central Banking* **1**(1), 55-93.

[60] Ha, J., and So, I. (2023). Which monetary shocks matter in small open economies? Evidence from Canada. *International Journal of Central Banking* **19**(2), 389-472.

[61] Hanson, S. G., and Stein, J. C. (2015). Monetary policy and long-term real rates. *Journal of Financial Economics* **115**(3), 429-448.

[62] Hausman, J., and Wongswan, J. (2011). Global asset prices and FOMC announcements. *Journal of International Money and Finance* **30**(3), 547-571.

[63] Havranek, T., and Rusnak, M. (2018). Transmission lags of monetary policy: A meta-analysis. *33rd issue (December 2013) of the International Journal of Central Banking*.

[64] Iacoviello, M., and Navarro, G. (2019). Foreign effects of higher US interest rates. *Journal of International Money and Finance* **95**, 232-250.

[65] Ibarra, R. (2016). How important is the credit channel in the transmission of monetary policy in Mexico? *Applied Economics* **48**(36), 3462-3484.

[66] Ibarra, R. (2023). The Yield Spread as a Predictor of Economic Activity in Mexico: The Role of the Term Premium. *Economía LACEA Journal* **22**(1), 153–174.

[67] Jaramillo, J., Pech Moreno, L. A., Ramírez, C., and Sanchez-Amador, D. (2019). Non-linear exchange rate pass-through in Mexico. (No. 2019-16). Banco de México Working Paper.

[68] Jarociński, M., and Karadi, P. (2020). Deconstructing monetary policy surprises—the role of information shocks. *American Economic Journal: Macroeconomics* **12**(2), 1-43.

[69] Jentsch, C., and Lunsford, K. G. (2019). Inference in structural vector autoregressions identified with external instruments. *Journal of the American Statistical Association* **114**(528), 1638–1650.

[70] Jentsch, C., and Lunsford, K. G. (2019). The dynamic effects of personal and corporate income tax changes in the United States: Comment. *American Economic Review* **109**(7), 2655-2678.

[71] Jordà, Ò. (2009). Simultaneous confidence regions for impulse responses. *The Review of Economics and Statistics* **91**(3), 629–647.

[72] Kerssenfischer, M. (2022). Information effects of euro area monetary policy. *Economics Letters* **216**, 1-5.

[73] Kim, D. H., and Singleton, K. J. (2012). Term structure models and the zero bound: an empirical investigation of Japanese yields. *Journal of Econometrics*, **170**(1), 32-49.

[74] Kim, S., and Lim, K. (2022). Effects of monetary policy shocks on exchange rate in emerging countries. *The World Economy* **45**(4), 1242-1261.

[75] Kim, S. H., Moon, S., and Velasco, C. (2017). Delayed overshooting: Is it an'80s puzzle? *Journal of Political Economy* **125**(5), 1570-1598.

[76] Kochen, F., and Sámano, D. (2016). Price-setting and exchange rate pass-through in the Mexican economy: Evidence from CPI micro data. (No. 2016-13). Banco de México Working Paper.

[77] Krippner, L. (2013). Measuring the stance of monetary policy in zero lower bound environments. *Economics Letters*, **118**(1), 135-138.

[78] Kuttner, K. N. (2001). Monetary policy surprises and interest rates: Evidence from the Fed funds futures market. *Journal of Monetary Economics* **47**(3), 523-544.

[79] Lindé, J. (2003). Monetary policy shocks and business cycle fluctuations in a small open economy: Sweden 1986-2002 (No. 153). Sveriges Riksbank Working Paper Series.

[80] Lombardi, D., Siklos, P., and St. Amand, S. (2018). A survey of the international evidence and lessons learned about unconventional monetary policies: Is a 'new normal'in our future? *Journal of Economic Surveys* **32**(5), 1229-1256.

[81] Maćkowiak, B. (2007). External shocks, US monetary policy and macroeconomic fluctuations in emerging markets. *Journal of Monetary Economics* **54**(8), 2512-2520.

[82] Mankiw, N. G. (2009). *Macroeconomics*. 7th Edition. Worth Publishers, New York.

[83] Mallick, S. K., and Sousa, R. M. (2012). Real effects of monetary policy in large emerging economies. *Macroeconomic Dynamics* **16**(S2), 190-212.

[84] Martínez, L., Sánchez, O., and Werner, A. (2001). Consideraciones sobre la conducción de la política monetaria y el mecanismo de transmisión en México (No. 2001-02). Banco de México Working Paper.

[85] Melosi, L. (2017). Signalling effects of monetary policy. *The Review of Economic Studies* **84**(2), 853-884.

[86] Mertens, K., and Ravn, M. O. (2013). The dynamic effects of personal and corporate income tax changes in the United States. *American economic review* **103**(4), 1212-1247.

[87] Mertens, K., and Ravn, M. O. (2019). The dynamic effects of personal and corporate income tax changes in the United States: Reply. *American Economic Review* **109**(7), 2679–2691.

[88] Minella, A. (2003). Monetary policy and inflation in Brazil (1975-2000): a VAR estimation. *Revista Brasileira de Economia* **57**, 605-635.

[89] Miranda-Agrippino, S., and Ricco, G. (2021). The transmission of monetary policy shocks. *American Economic Journal: Macroeconomics* **13**(3), 74-107.

[90] Mishkin, F. S. (1995). Symposium on the monetary transmission mechanism. *Journal of Economic perspectives* **9**(4), 3-10.

[91] Moessner, R., Jansen, D.-J., and de Hann, J. (2017). Communication about future policy rates in theory and practice: a survey. *Journal of Economic Surveys* **31**(3), 678-711.

[92] Morsink, J., and Bayoumi, T. (2001). A peek inside the black box: the monetary transmission mechanism in Japan. *IMF Staff Papers* **48**(1), 22-57.

[93] Murno, J. (2025). EMTA Survey: Second Quarter 2025 Emerging Markets Debt Trading at US$1.464 Trillion. Emerging Markets Traders Association, Press Release, September 29, 2025.

[94] Nakamura, E., and Steinsson, J. (2018). High-frequency identification of monetary non-neutrality: the information effect. *The Quarterly Journal of Economics* **133**(3), 1283-1330.

[95] Nguyen, T. M. L. (2020). Output effects of monetary policy in emerging and developing countries: Evidence from a meta-analysis. *Emerging Markets Finance and Trade* **56**(1), 68-85.

[96] Ormaechea, M. S. A., and Fernandez, M. D. C. (2011). The monetary transmission in dollarized and non-dollarized economies: the cases of Chile, New Zealand, Peru and Uruguay. *International Monetary Fund*.

[97] Otero, J. D. Q. (2015). Impactos de la política monetaria y canales de transmisión en países de América Latina con esquema de inflación objetivo. *Ensayos sobre Política Económica* **33**(76), 61–75.

[98] Orphanides, A. (2001). Monetary Policy Rules Based on Real-Time Data. *American Economic Review*, **91**(4), 964–985.

[99] Racette, D., and Raynauld, J. (1992). Canadian monetary policy: will the checklist approach ever get us to price stability? *Canadian Journal of Economics*, 819-838.

[100] Ramey, V. A. (1993). How important is the credit channel in the transmission of monetary policy? *Carnegie-Rochester Conference Series on Public Policy* **39**(1), 1-45.

[101] Ramos-Francia, M., and Torres, A. (2008). Inflation dynamics in Mexico: a characterization using the new Phillips curve. *The North American Journal of Economics and Finance* **19**(3), 274-289.

[102] Rogers, J. H., Scotti, C., and Wright, J. H. (2014). Evaluating asset-market effects of unconventional monetary policy: a multi-country review. *Economic Policy* **29**(80), 749-799.

[103] Romer, C. D., and Romer, D. H. (2000). Federal Reserve information and the behavior of interest rates. *American Economic Review* **90**(3), 429-457.

[104] Romer, C. D., and Romer, D. H. (2004). A new measure of monetary shocks: Derivation and implications. *American Economic Review* **94**(4), 1055-1084.

[105] Rosa, C. (2011). The high-frequency response of exchange rates to monetary policy actions and statements. *Journal of Banking & Finance* **35**(2), 35(2), 478-489.

[106] Rusnák, M., Havranek, T., and Horváth, R. (2013). How to solve the price puzzle? A meta-analysis. *Journal of Money, Credit and Banking* **45**(1), 37-70.

[107] Sidaoui, J. and Ramos-Francia, M. (2008). The monetary transmission mechanism in Mexico: recent developments. BIS Papers, **35**, 363-394.

[108] Sims, C. A. (1980). Macroeconomics and reality. *Econometrica: Journal of the Econometric Society* **48**, 1-48.

[109] Sims, C. A. (1992). Interpreting the macroeconomic time series facts: The effects of monetary policy. *European Economic Review* **36**(5), 975-1000.

[110] Sims, C. A., Stock, J. H., & Watson, M. W. (1990). Inference in Linear Time Series Models with Some Unit Roots. *Econometrica*, **58**(1), 113–144.

[111] Sims, C. A., and Zha, T. (2006). Does monetary policy generate recessions? *Macroeconomic Dynamics* **10**(2), 231-272.

[112] Solís, P. (2023a). Does the exchange rate respond to monetary policy in Mexico? Solving an exchange rate puzzle in emerging markets. *Journal of Money, Credit and Banking* **55**(8), 2093-2113.

[113] Solís, P. (2023b). Do central bank words matter in emerging markets? Evidence from Mexico. *Journal of Macroeconomics* **78**, 1-16.

[114] Soosalu, M. (2024). Monetary policy transmission in Canada - A high frequency identification approach. *The B.E. Journal of Macroeconomics* **24**(2), 781-811.

[115] Stock, J. H., and Watson, M. W. (2002). Macroeconomic forecasting using diffusion indexes. *Journal of Business & Economic Statistics* **20**(2), 147-162.

[116] Stock, J. H., and Watson, M. W. (2012). Disentangling the Channels of the 2007-09 Recession. *Brookings Papers on Economic Activity* **42**(1), 81-135.

[117] Stock, J. H., and Watson, M. W. (2018). Identification and estimation of dynamic causal effects in macroeconomics using external instruments. *The Economic Journal* **128**(610), 917-948.

[118] Stock, J. H., Wright, J. H., and Yogo, M. (2002). A survey of weak instruments and weak identification in generalized method of moments. *Journal of Business and Economic Statistics* **20**(4), 518-529.

[119] Swanson, E. T., and Williams, J. C. (2014). Measuring the effect of the zero lower bound on medium-and longer-term interest rates. *American Economic Review* **104**(10), 3154-3185.

[120] Taylor, J. B. (1995). The monetary transmission mechanism: an empirical framework. *Journal of Economic Perspectives* **9**(4), 11-26.

[121] Trenkler, C., and Weber, E. (2020). Identifying shocks to business cycles with asynchronous propagation. *Empirical Economics* **58**(4), 1815–1836.

[122] Trung, N. B. (2022). Output fluctuations and portfolio flows to emerging economies: The role of monetary uncertainty. *International Finance* **25**(3), 285-295.

[123] Uhlig, H. (2005). What are the effects of monetary policy on output? Results from an agnostic identification procedure. *Journal of Monetary Economics* **52**(2), 381-419.

[124] Uribe, M., and Yue, V. Z. (2006). Country spreads and emerging countries: Who drives whom? *Journal of international Economics* **69**(1), 69(1), 6-36.

[125] Vicondoa, A. (2019). Monetary news in the United States and business cycles in emerging economies. *Journal of International Economics* **117**, 79-90.

[126] World Bank (2025). World Development Indicators. Imports of goods and services (current US$), and related series.

[127] Wright, J. H. (2012). What does monetary policy do to long-term interest rates at the zero lower bound? *The Economic Journal* **122**(564), F447-F466.

[128] Wu, J. C., and Xia, F. D. (2016). Measuring the macroeconomic impact of monetary policy at the zero lower bound. *Journal of Money, Credit and Banking* **48**(2-3), 253-291.

A Appendix: Identification

Following Gertler and Karadi (2015), we partition the vector of reduced form residuals as $\mathbf{u}_t = [u_t^p \; \mathbf{u}_t^{q\prime}]' = [u_{1t} \; \mathbf{u}_{2t}']'$, and the corresponding matrix of structural coefficients as

$$\mathbf{S} = [\mathbf{s} \; \mathbf{S}^q] = [\mathbf{S}_1 \; \mathbf{S}_2] = \begin{bmatrix} s_{11} & \mathbf{s}_{12} \\ \mathbf{s}_{21} & \mathbf{s}_{22} \end{bmatrix},$$

and the reduced form variance-covariance matrix as

$$\boldsymbol{\Sigma} = \begin{bmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{bmatrix}.$$

Here, s^p is identified up to a sign convention and can be obtained by the following closed form solution

$$(s^p)^2 = s_{11}^2 = \boldsymbol{\Sigma}_{11} - \mathbf{s}_{12}\mathbf{s}_{12}',$$

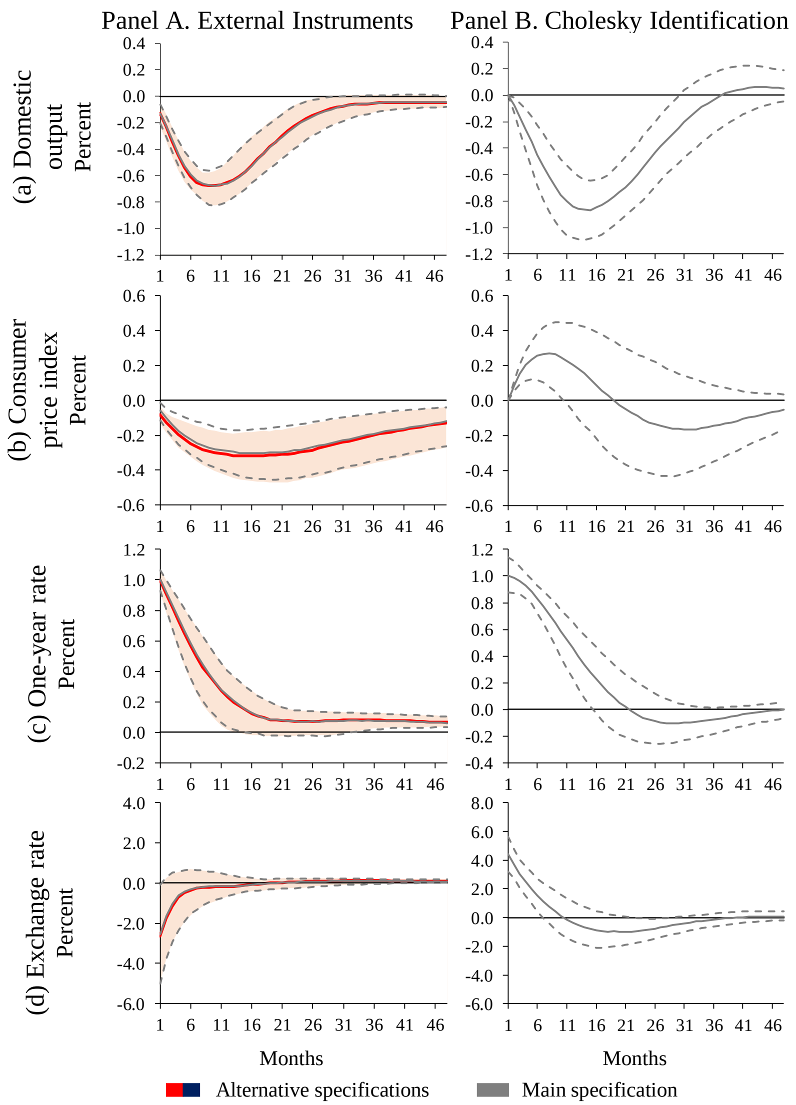
where

$$\mathbf{s}_{12}\mathbf{s}_{12}' = (\boldsymbol{\Sigma}_{21} - \frac{\mathbf{s}_{21}}{s_{11}}\boldsymbol{\Sigma}_{11})' Q^{-1} (\boldsymbol{\Sigma}_{21} - \frac{\mathbf{s}_{21}}{s_{11}}\boldsymbol{\Sigma}_{11}),$$

with

$$Q = \frac{\mathbf{s}_{21}}{s_{11}}\boldsymbol{\Sigma}_{11}\frac{\mathbf{s}_{21}'}{s_{11}} - (\boldsymbol{\Sigma}_{21}\frac{\mathbf{s}_{21}'}{s_{11}} + \frac{\mathbf{s}_{21}'}{s_{11}}\boldsymbol{\Sigma}_{21}') + \boldsymbol{\Sigma}_{22}.$$

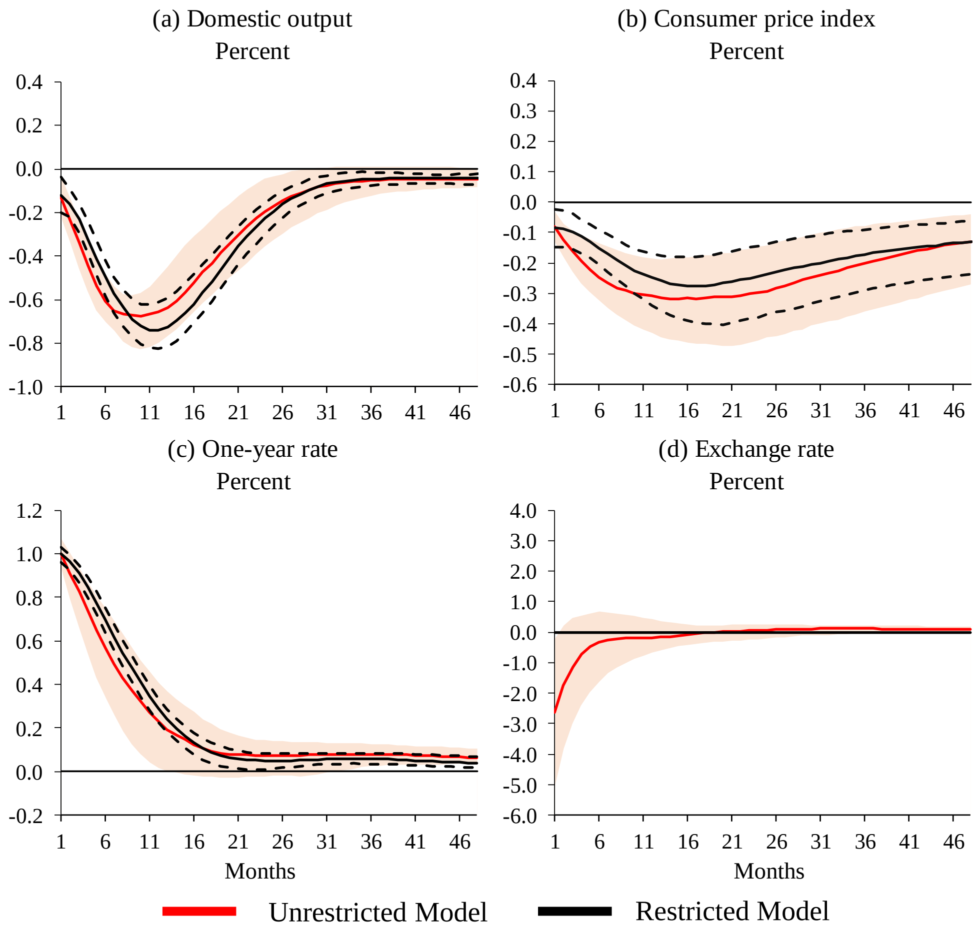
B Appendix: Supplemental Results



First-stage regression: F: 10.73; Robust F: 12.47; R^2: 9.36%; $\bar{R}^2$: 8.48%

Figure B.1: Impulse Response Functions to a Monetary Policy Shock by using an alternative Monetary Policy Surprise computed within a 50-Minute Window

Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.



First-stage regression: F: 10.73; Robust F: 12.47; R^2: 9.36%; $\bar{R}^2$: 8.48%

Figure B.2: Impulse Response Functions to a Monetary Policy Shock estimated using the Unrestricted and the Restricted Model with the Exogenized Exchange Rate: Specification using an alternative Monetary Policy Surprise computed within a 50-Minute Window

Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics and the R-squared values correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.

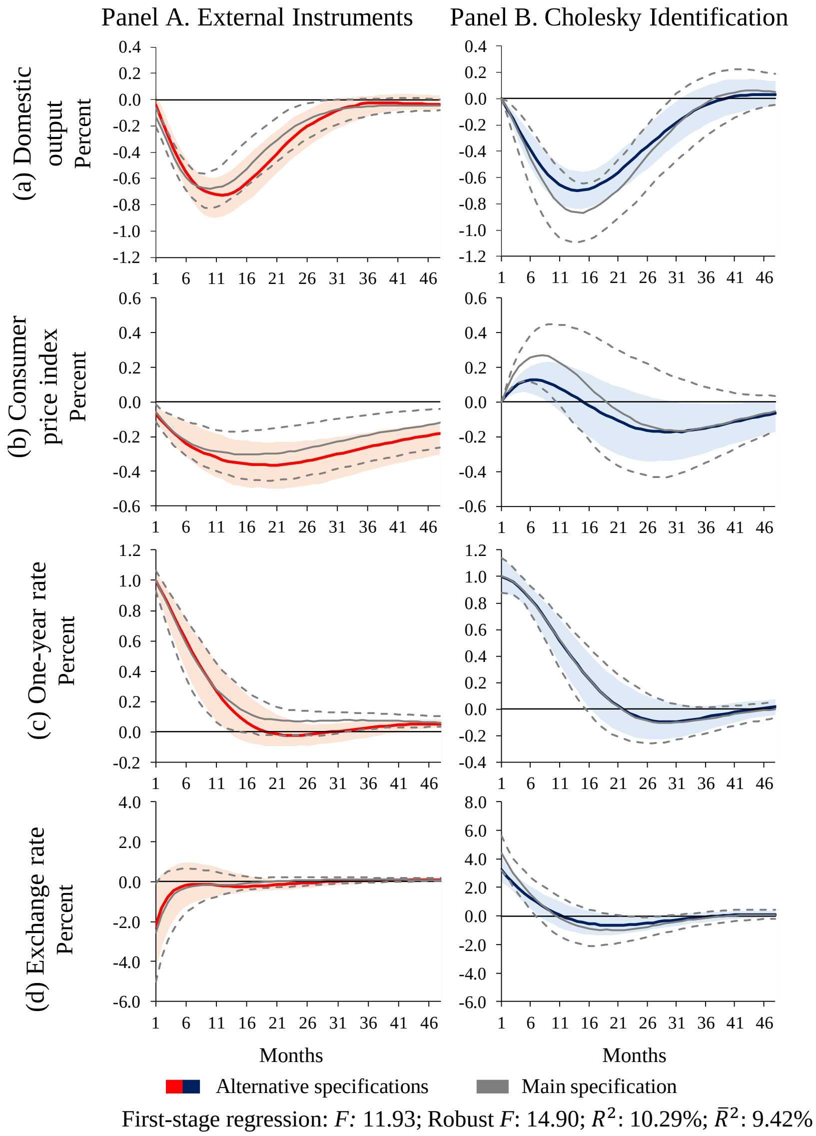
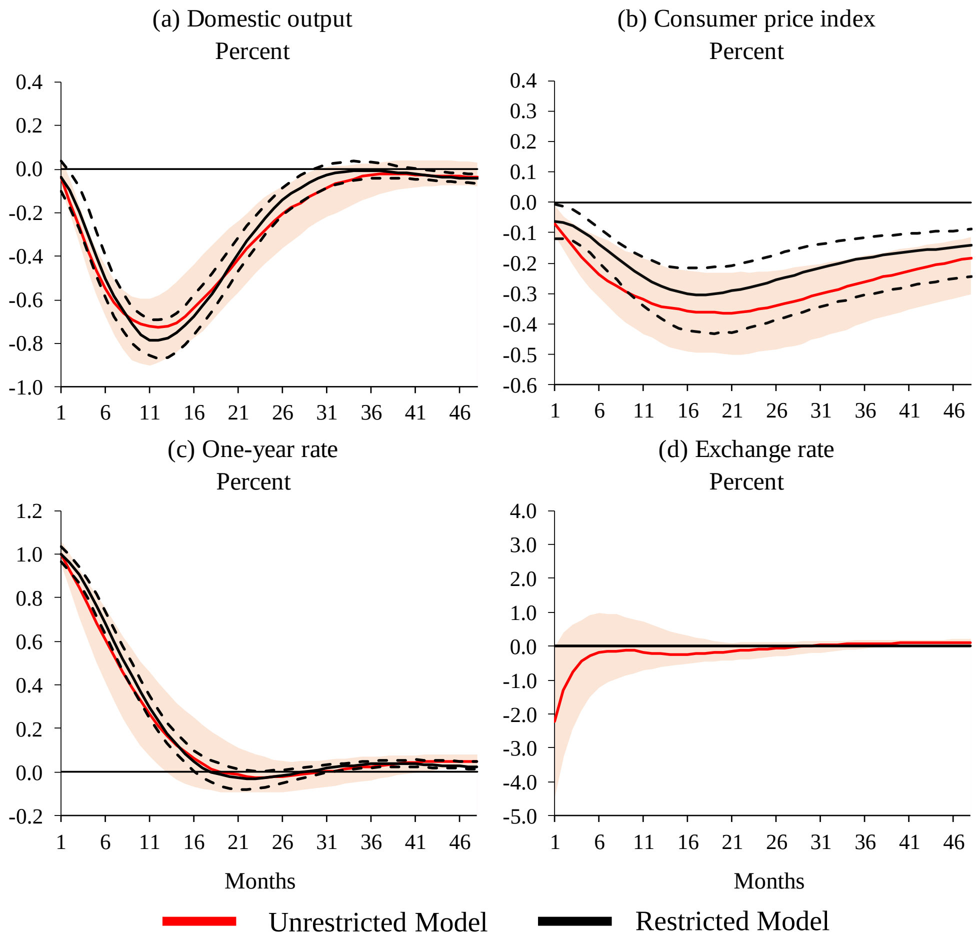


Figure B.3: Impulse Response Functions by using the US 2-Year Government Bond Yield instead of the Shadow Interest Rate

Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.



First-stage regression: F: 11.93; Robust F: 14.90; R^2: 10.29%; $\bar{R}^2$: 9.42%

Figure B.4: Impulse Response Functions to a Monetary Policy Shock estimated using the Unrestricted and the Restricted Model with the Exogenized Exchange Rate: Specification using the US 2-Year Government Bond Yield instead of the Shadow Interest Rate

Notes:Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics and the R-squared values correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.

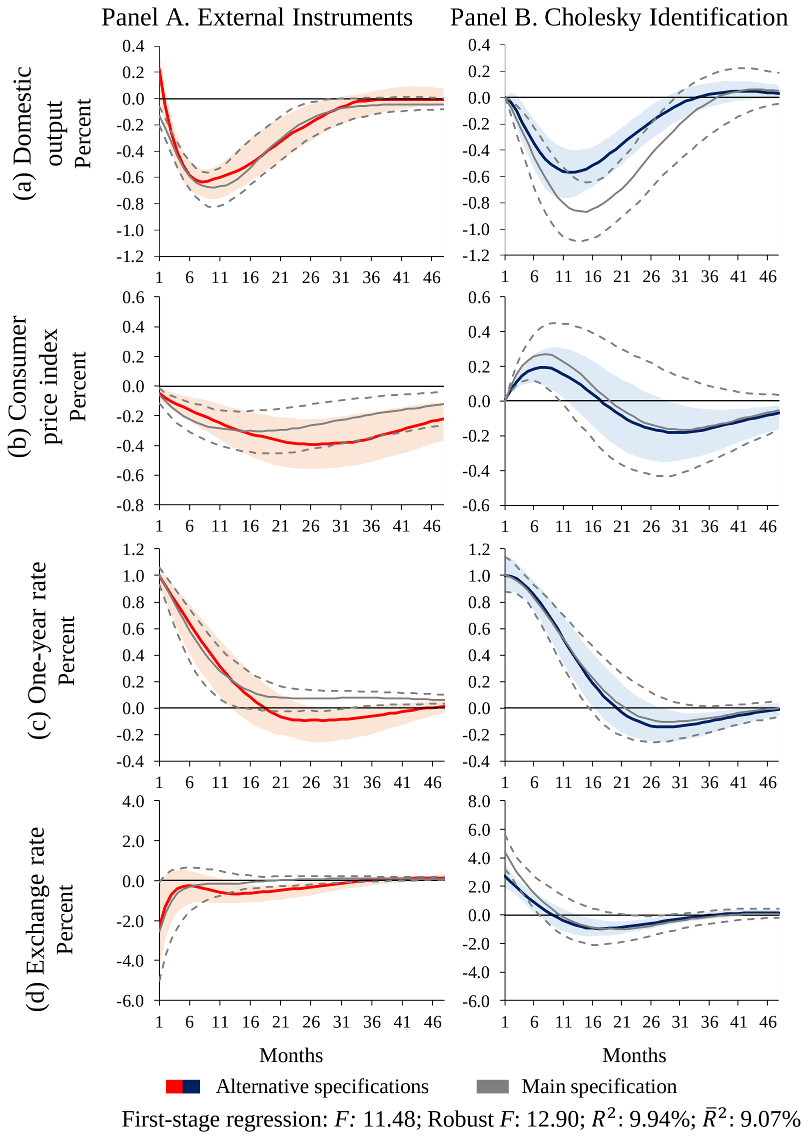
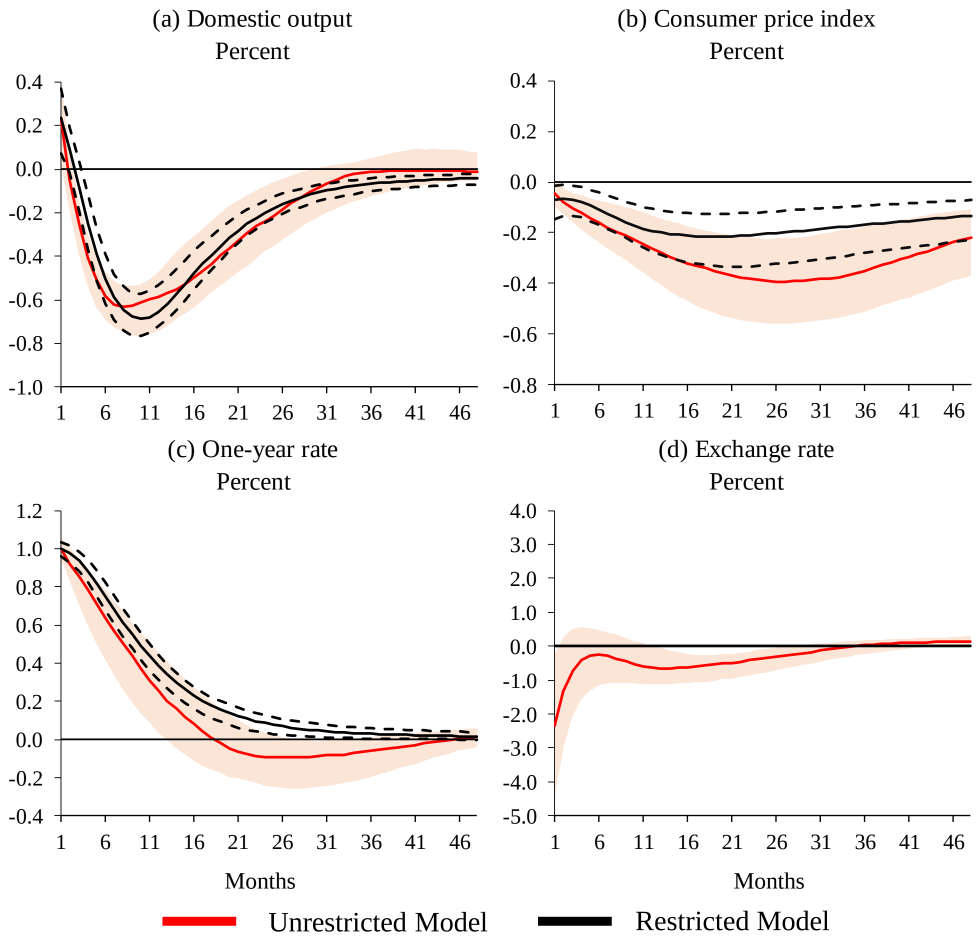


Figure B.5: Impulse Response Functions by using the Industrial Production Index as an Alternative Measure of Economic Activity

Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.



First-stage regression: F: 11.48; Robust F: 12.90; R^2: 9.94%; $\bar{R}^2$: 9.07%

Figure B.6: Impulse Response Functions to a Monetary Policy Shock estimated using the Unrestricted and the Restricted Model with the Exogenized Exchange Rate: Specification using the Industrial Production Index as an Alternative Measures of Economic Activity

Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics and the R-squared values correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.

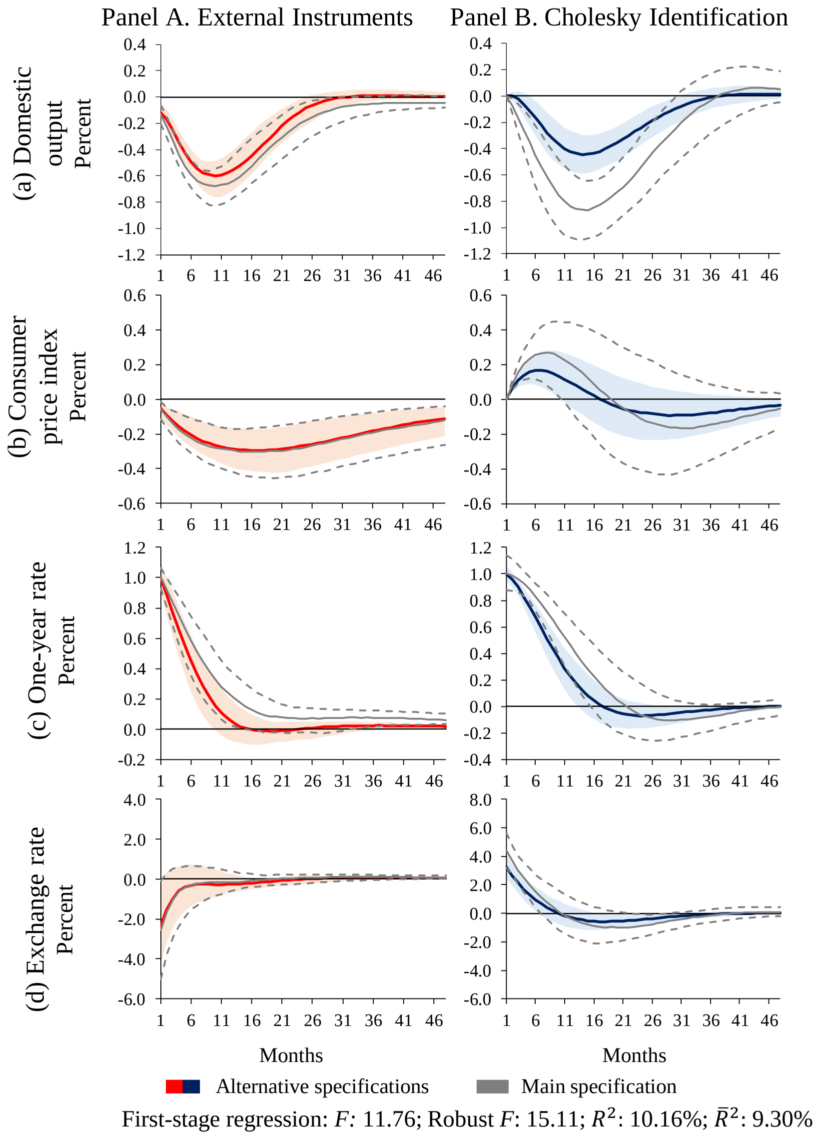
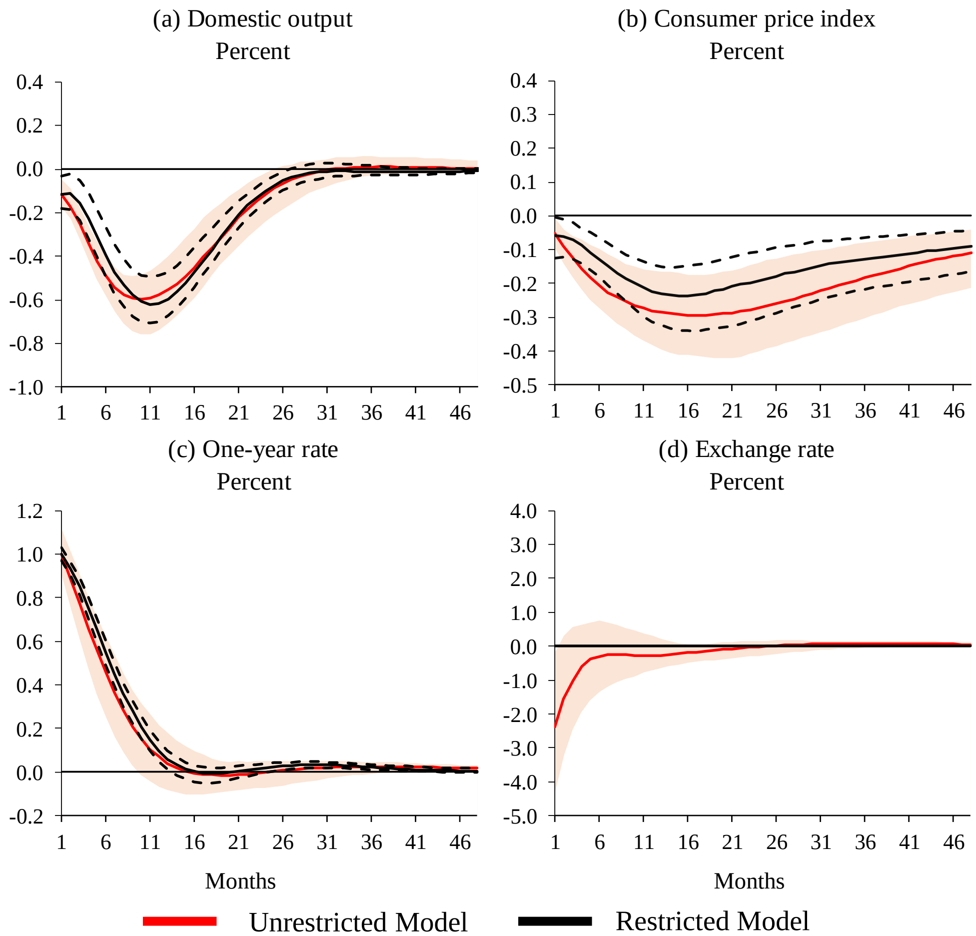


Figure B.7: Impulse Response Functions by using the 2-Year Government Bond Yield as the Monetary Policy Indicator

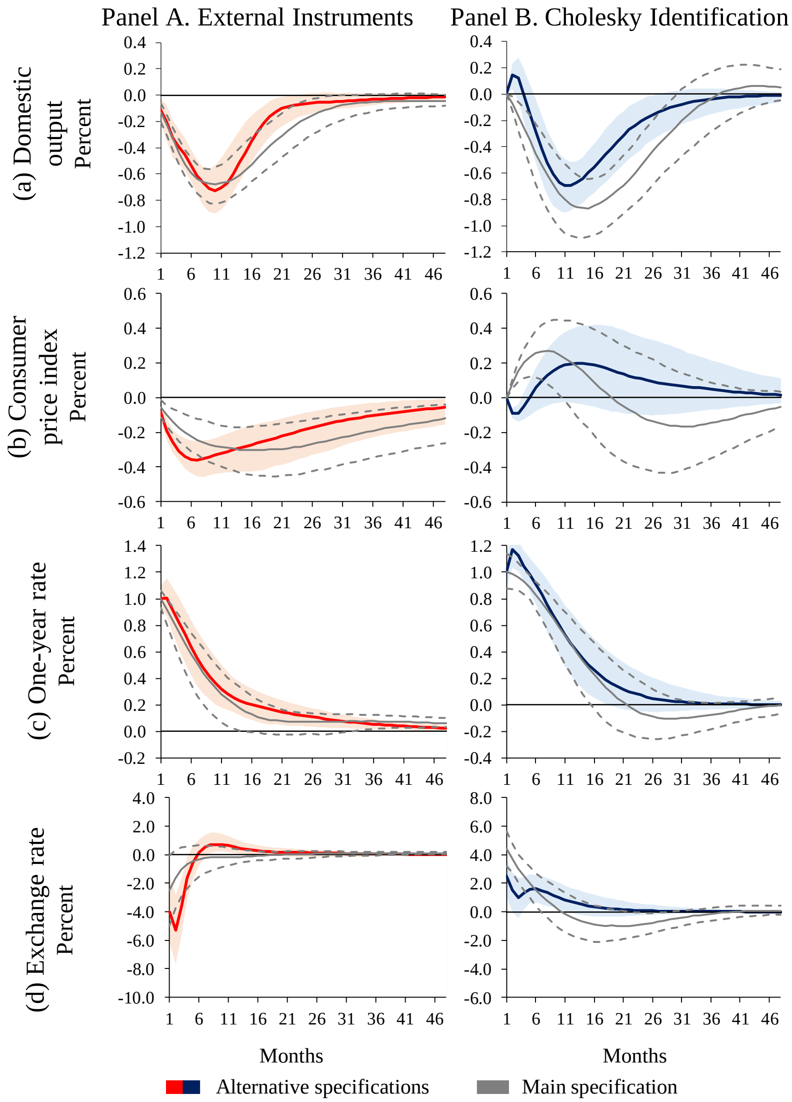
Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics correspond to the first stage regression of the two-year bond rate residual on the 3-month swap rate surprises.



First-stage regression: F: 11.76; Robust F: 15.11; R^2: 10.16%; $\bar{R}^2$: 9.30%

Figure B.8: Impulse Response Functions to a Monetary Policy Shock estimated using the Unrestricted and the Restricted Model with the Exogenized Exchange Rate: Specification using the 2-Year Government Bond Yield as the Monetary Policy Indicator

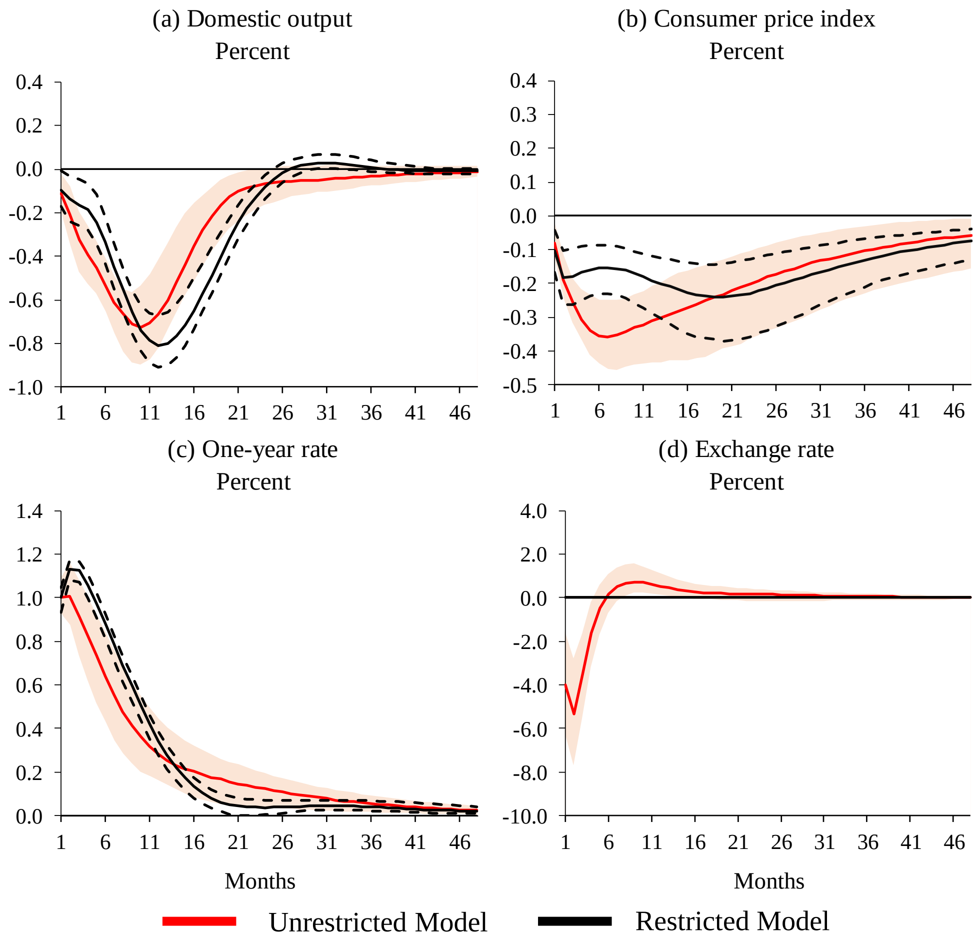
Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics and the R-squared values correspond to the first stage regression of the two-year bond rate residual on the 3-month swap rate surprises.



First-stage regression: *F*: 9.40; Robust *F*: 9.51; R^2: 8.29%; $\bar{R}^2$: 7.41%

Figure B.9: Impulse Response Functions to a Monetary Policy Shock: VAR Model with 2 Lags

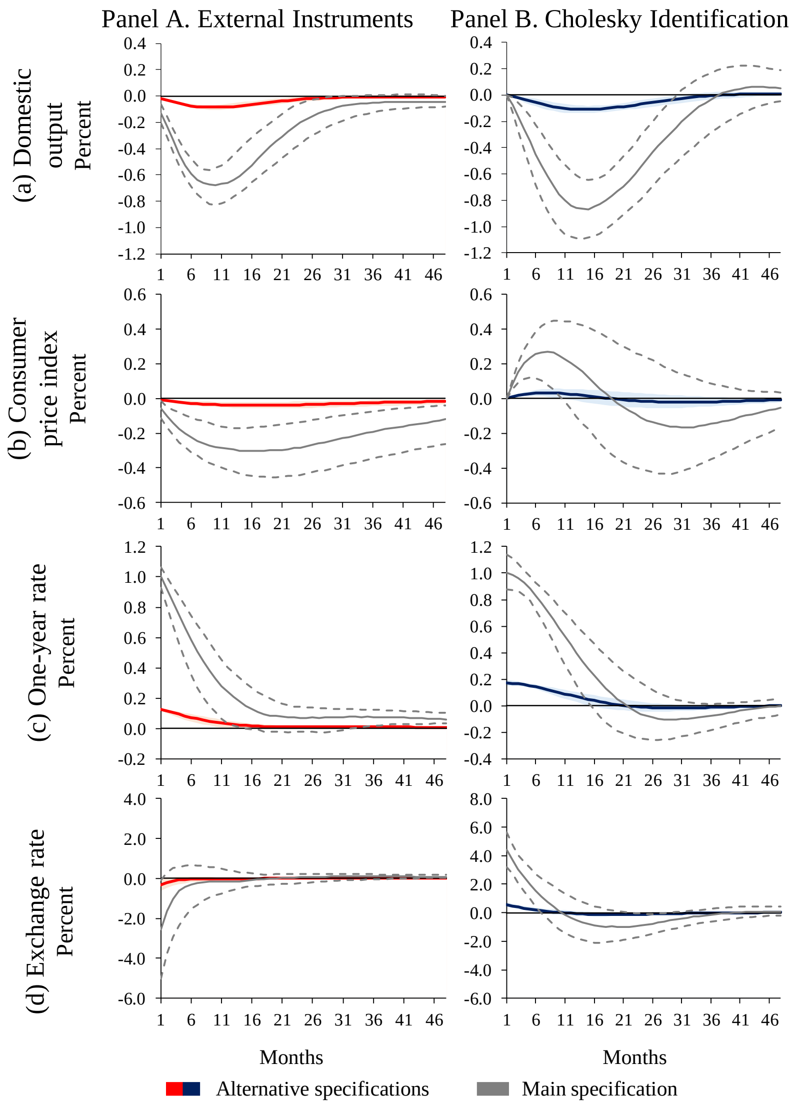
Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.



First-stage regression: F: 9.40; Robust F: 9.51; R^2: 8.29%; $\bar{R}^2$: 7.41%

Figure B.10: Impulse Response Functions to a Monetary Policy Shock estimated using the Unrestricted and the Restricted Model with the Exogenized Exchange Rate: VAR Model with 2 Lags

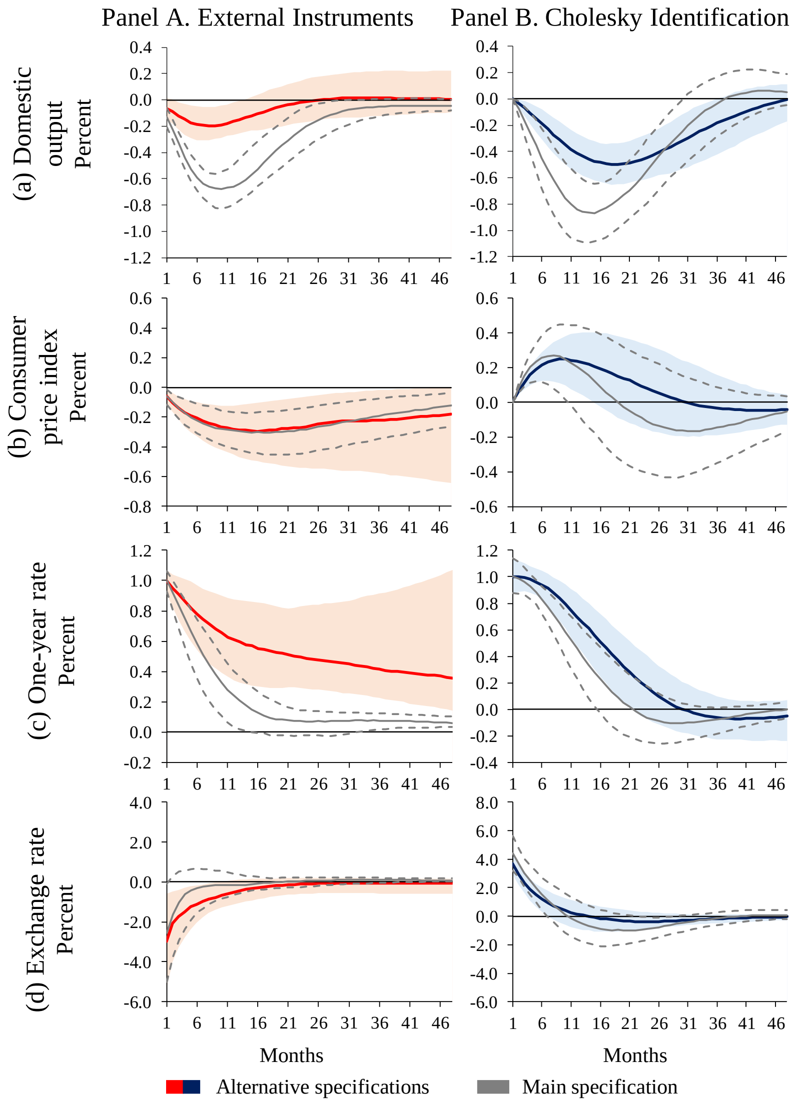
Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics and the R-squared values correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.



First-stage regression: *F*: 10.78; Robust *F*: 12.27; R^2: 9.39%; $\bar{R}^2$: 8.52%

Figure B.11: Impulse Response Functions without Standardization.

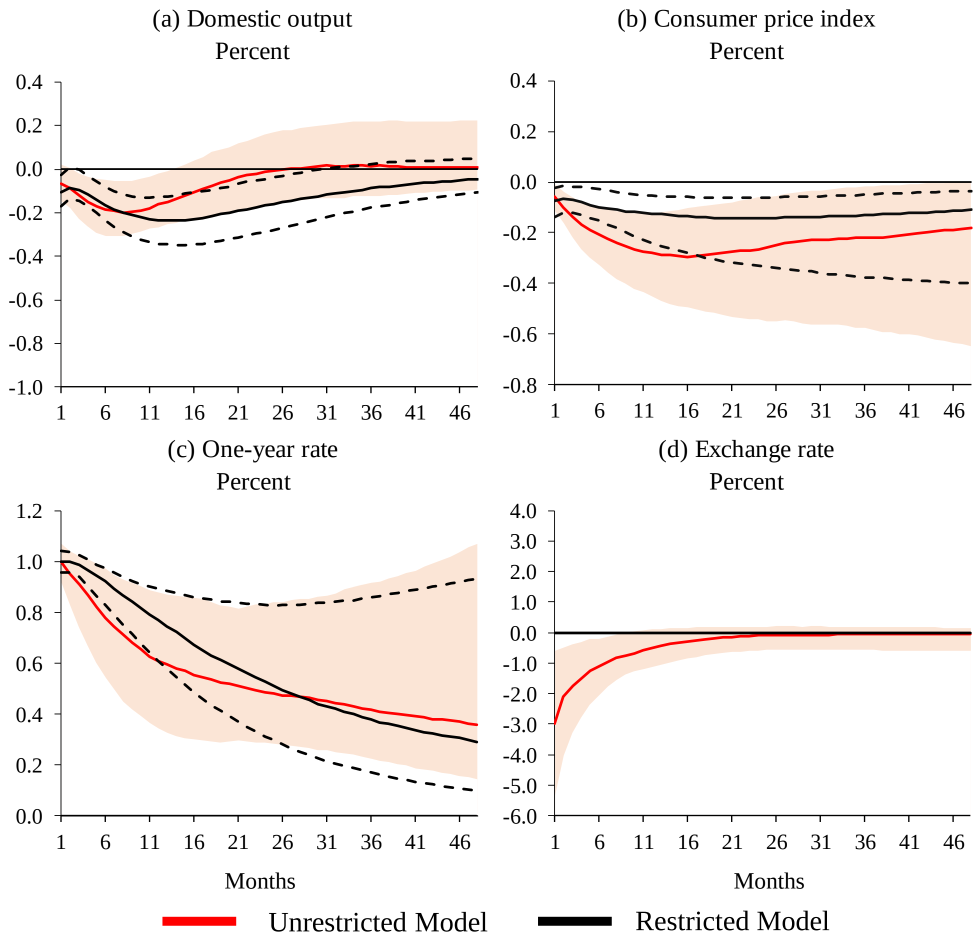
Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.



First-stage regression: F: 11.67; Robust F: 13.84; R^2: 10.09%; $\bar{R}^2$: 9.22%

Figure B.12: Impulse Response Functions by adding the US Excess Bond Premium

Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.



First-stage regression: F: 11.67; Robust F: 13.84; R^2: 10.09%; $\bar{R}^2$: 9.22%

Figure B.13: Impulse Response Functions to a Monetary Policy Shock estimated using the Unrestricted and the Restricted Model with the Exogenized Exchange Rate: Specification adding the US Excess Bond Premium

Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics and the R-squared values correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.

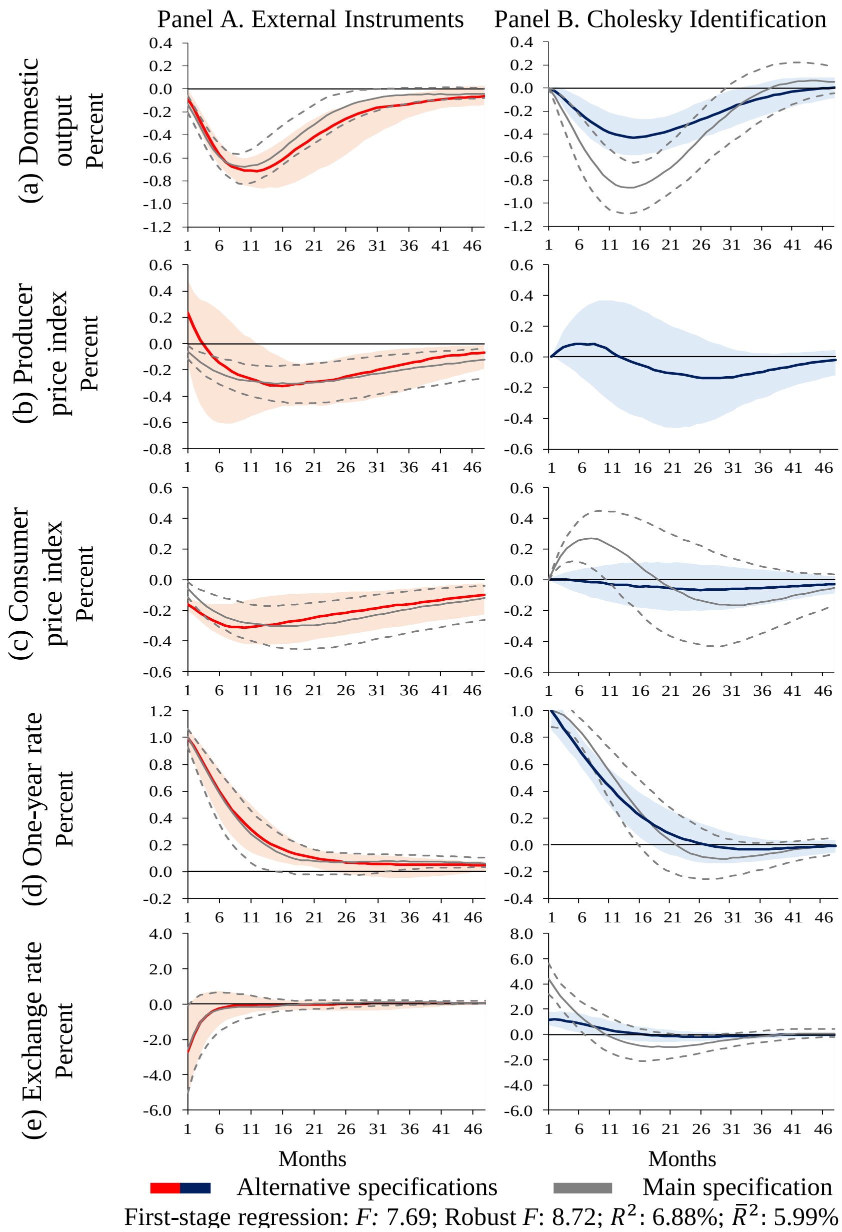
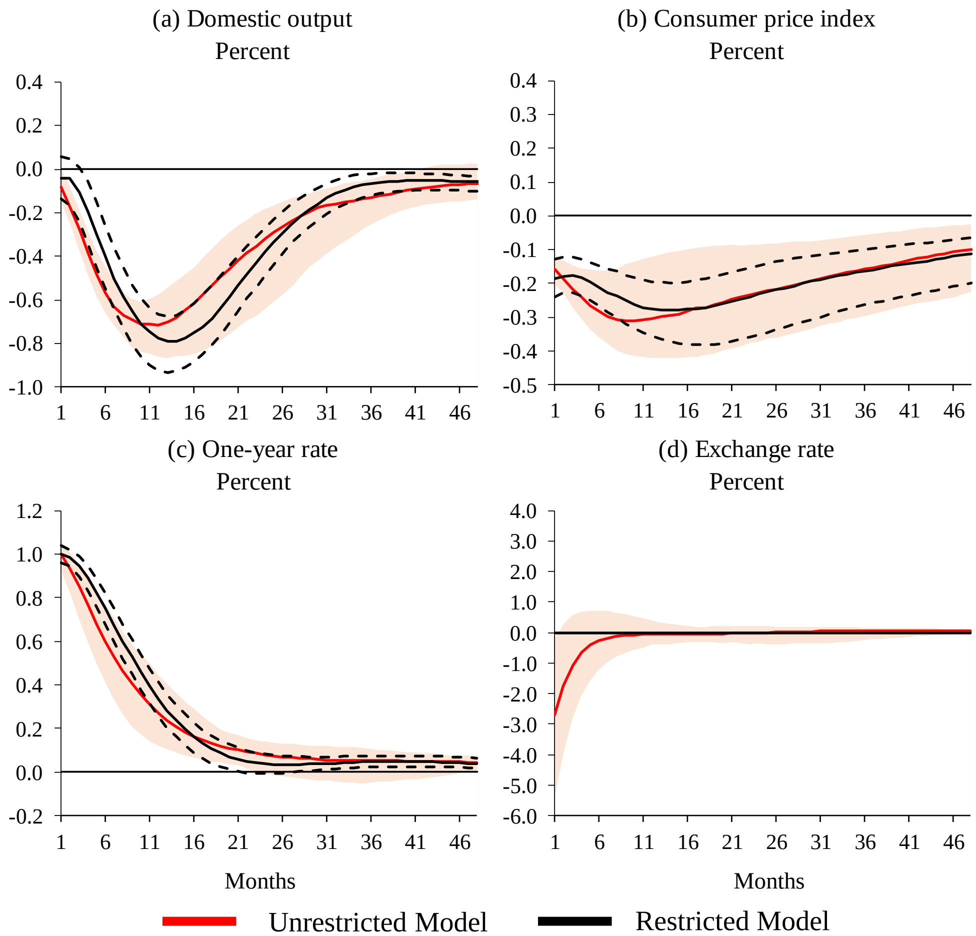


Figure B.14: Impulse Response Functions by adding the Producer Price Index

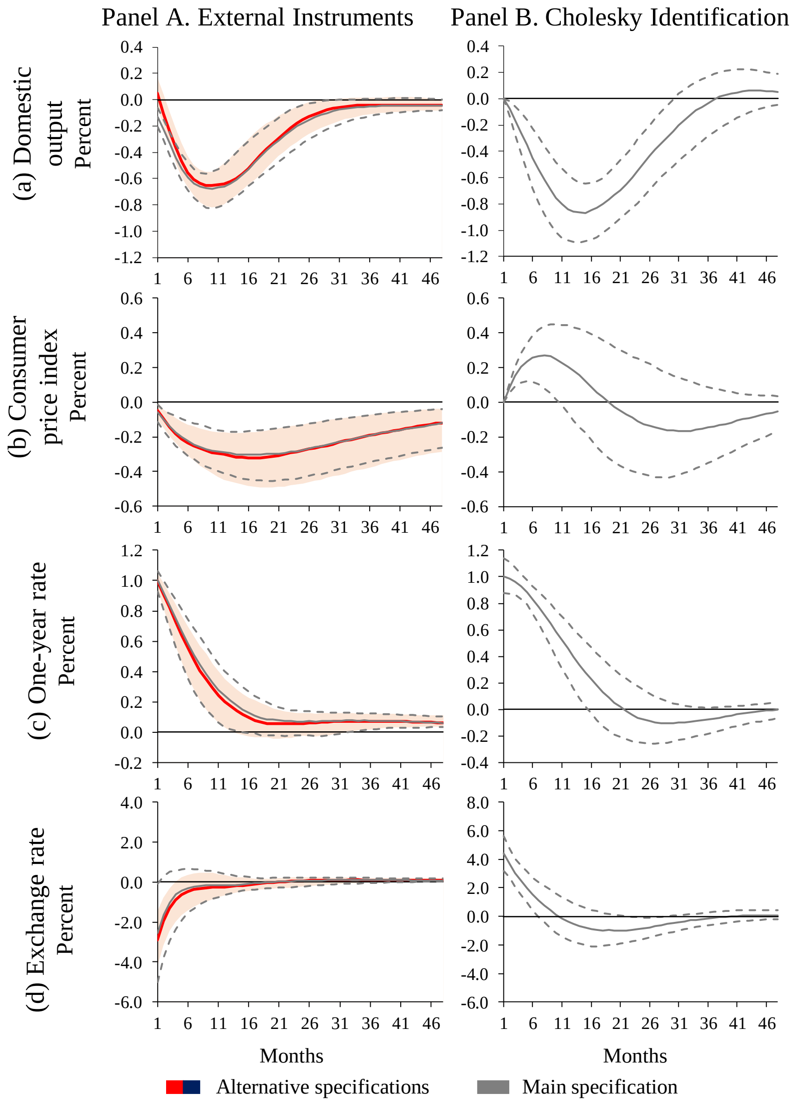
Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.



First-stage regression: *F:* 7.69; Robust *F*: 8.72; R^2: 6.88%; $\bar{R}^2$: 5.99%

Figure B.15: Impulse Response Functions to a Monetary Policy Shock estimated using the Unrestricted and the Restricted Model with the Exogenized Exchange Rate: Specification adding the Producer Price Index

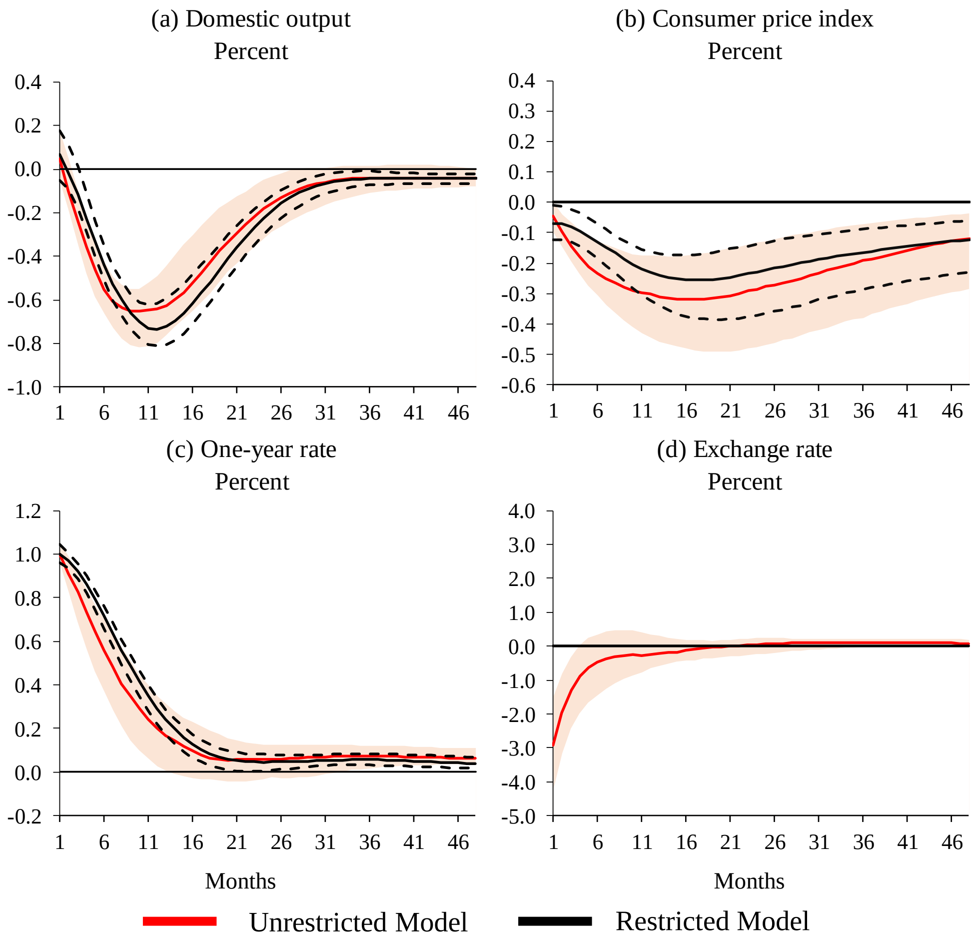
Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics and the R-squared values correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises.



First-stage regression: *F:* 12.33; Robust *F*: 25.64; R^2: 10.60%; $\bar{R}^2$: 9.74%

Figure B.16: Impulse Response Functions by using Monetary Policy Surprises purged from Information Effects as an External Instrument

Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises purged from information effects.



First-stage regression: F: 12.33; Robust F: 25.64; R^2: 10.60%; $\bar{R}^2$: 9.74%

Figure B.17: Impulse Response Functions to a Monetary Policy Shock estimated using the Unrestricted and the Restricted Model with the Exogenized Exchange Rate: Specification using Monetary Policy Surprises purged from Information Effects as an External Instrument

Notes: Responses are presented for a 48-month horizon along with 68 percent confidence intervals, computed using moving block bootstrap. The F-statistics and the R-squared values correspond to the first stage regression of the one-year bond rate residual on the 3-month swap rate surprises purged from information effects.

C Appendix: Exogenizing a variable

This appendix illustrates the procedure used to turn off the exchange rate propagation channel in Section 3.2. Consider the following VAR(1):

$$x_{1,t} = \alpha_1 + \beta_{11} x_{1,t-1} + \beta_{12} x_{2,t-1} + \beta_{13} x_{3,t-1} + u_{1,t}, \tag{C.1}$$

$$x_{2,t} = \alpha_2 + \beta_{21} x_{1,t-1} + \beta_{22} x_{2,t-1} + \beta_{23} x_{3,t-1} + u_{2,t}, \tag{C.2}$$

$$x_{3,t} = \alpha_3 + \beta_{31} x_{1,t-1} + \beta_{32} x_{2,t-1} + \beta_{33} x_{3,t-1} + u_{3,t}, \tag{C.3}$$

where $u_t = (u_{1,t}, u_{2,t}, u_{3,t})'$ are reduced-form innovations. For simplicity, the illustration is based on a three-variable with one lag model, although the procedure extends directly to larger systems.

To exogenize $x_{3,t}$, remove its equation from the endogenous block and treat its lags as exogenous regressors in the remaining equations, as follows:

$$x_{1,t} = \alpha_1 + \beta_{11} x_{1,t-1} + \beta_{12} x_{2,t-1} + \beta_{13} x_{3,t-1} + u_{1,t}, \tag{C.4}$$

$$x_{2,t} = \alpha_2 + \beta_{21} x_{1,t-1} + \beta_{22} x_{2,t-1} + \beta_{23} x_{3,t-1} + u_{2,t}. \tag{C.5}$$

This procedure generates a VAR whose equations have the same estimated parameters as in the original system, but whose impulse responses can differ because any transmission that operates through $x_{3,t}$ is now blocked. Consequently, comparing the impulse response functions from (C.1)–(C.3) with those from (C.4)–(C.5) provides a measure of the importance of this channel in the transmission mechanism.